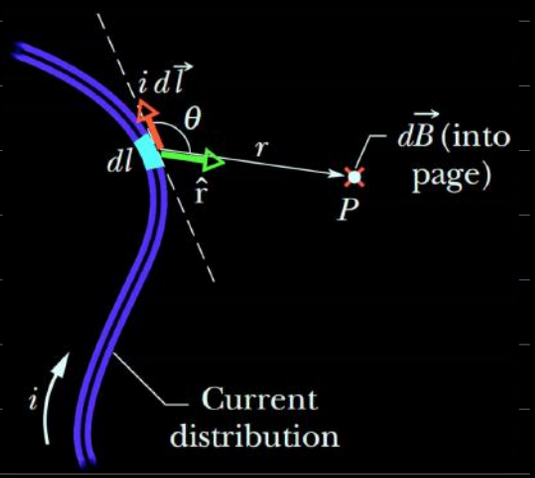
Biot & Savart Law: ($\vec{B}$ Due To Current Element):

The magnetic field due to a current element $i\vec{dl}$, the magnetic field at any point having a position vector $\vec{r}$ wrt current element is given as:

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\,d\vec{l}\times\vec{r}}{r^3}}$$

, Here by current element we mean a directed vector having magnitude idl and direction along local current flow (i.e against drift velocity of electrons).

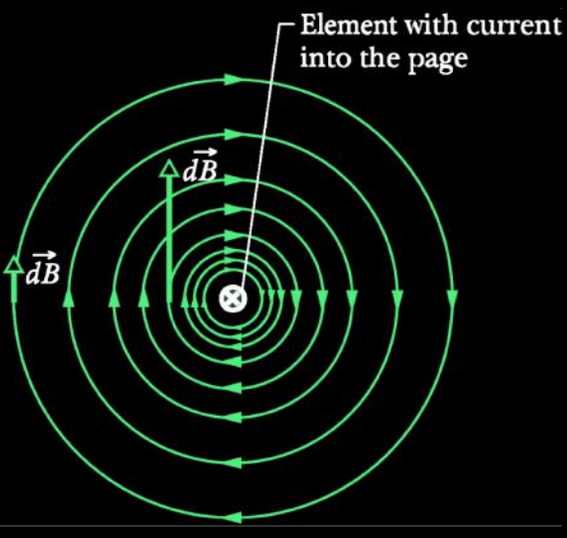


→ Here $\mu_0/4\pi$ is a constant having a value 10^{-7} Tm/A, μ_0 is called permeability of free space, $\mu_0 = 4\pi\times10^{-7} \approx 1.26\times10^{-6}$ Tm/A.

→ Magnitude of field due to current element can be written as

$$\boxed{dB = \frac{\mu_0}{4\pi}\frac{idl\sin\theta}{r^2}}$$

→ It can be clearly seen that a current element produces a tangential magnetic field pattern in form of concentric circles, thus magnetic field lines due to a current element form closed lines.



→ If we consider a Gaussian surface in space, the net magnetic flux through it due to current element must be zero because every field line is forming a closed loop. (Thus every field line entering the Gaussian surface must also exit somewhere). This fact can be used to deduce the Gauss law for Magnetism, i.e $\oint \vec{B}\cdot d\vec{S} = 0$.

Gauss Law For Magnetism: In the previous section we showed that for a current element $\oint \vec{B}\cdot d\vec{S} = 0$, now any configuration of currents in space can be thought to be composed of several current elements and for each one of them $\oint \vec{B}\cdot d\vec{S} = 0$

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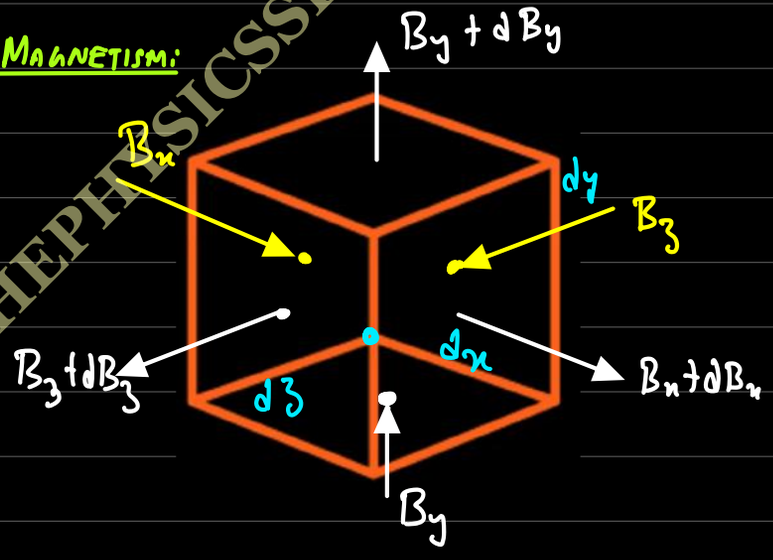
thus even for arbitrary configurations of currents in space we can say that $\oint \vec{B} \cdot d\vec{S} = 0$. Or in words we can say that net magnetic flux through any closed surface is always zero. This is known as Gauss' law for magnetism and it is one of the fundamental equations of Maxwell:

$$\boxed{\oint \vec{B} \cdot d\vec{S} = 0}$$

→ This also proves that magnetic field lines always form closed loops, because if any field line were not to form a closed loop, we could enclose its end in a small Gaussian surface causing a non-zero flux.

Differential Form Of Gauss Law For Magnetism:

Consider a differential Gaussian Cuboid of dimensions $dx \times dy \times dz$. The components of fields associated with this cube's various faces are shown. Now applying Gauss Law for Magnetism:



$$(B_x + dB_x)\,dy\,dz - B_x(dy\,dz) + (B_y + dB_y)\,dx\,dz - B_y\,dx\,dz + (B_z + dB_z)\,dx\,dy - B_z\,dx\,dy = 0$$

Thus $dB_x\,dy\,dz + dB_y\,dx\,dz + dB_z\,dx\,dy = 0$, Now dividing by $dx\,dy\,dz$:

$$\frac{dB_x}{dx} + \frac{dB_y}{dy} + \frac{dB_z}{dz} = 0$$, but since B_x, B_y and B_z are multivariable functions thus here we actually mean partial derivatives. Thus:

$$\boxed{\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0} \quad \text{or} \quad \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

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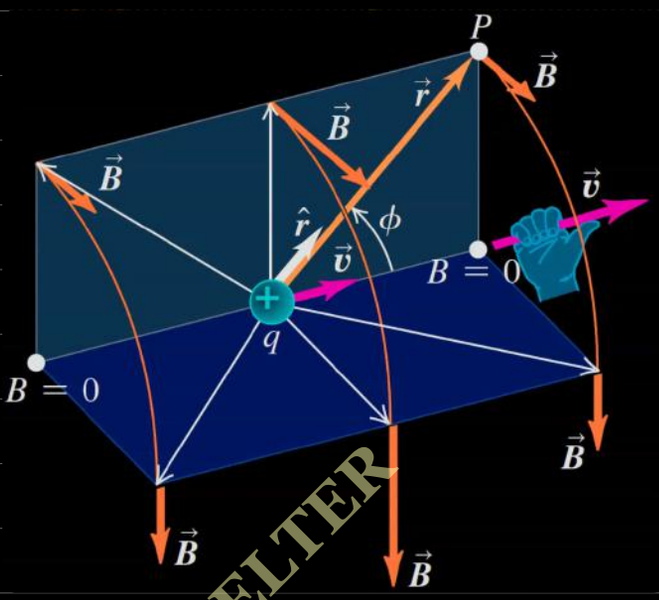
Magnetic Field Due To A Moving Point Charge:



Recall that magnetic field due to a current element $id\vec{l}$ was given as: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \vec{r}}{r^3}$ now let dq be the moving charge within the element moving with a drift velocity $\vec{v}_d$ which would cross it in a time dt (i.e cross the length dl in a time dt) then we can also write the field as:

$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \left(\frac{dq}{dt}\right) dl\, \hat{v}_d \times \frac{\vec{r}}{r^3}$$

$$\Rightarrow d\vec{B} = \frac{\mu_0}{4\pi} dq \frac{\vec{v}_d \times \vec{r}}{r^3} \quad \left(\because \frac{dl}{dt}\hat{v}_d = \vec{v}_d\right)$$

Similarly instead of an elemental charge we had a finite point charge moving with a velocity $\vec{v}$ then:

$$\boxed{\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \vec{r}}{r^3}} \quad (1)$$

→ Just like with a current element, magnetic field lines due to a point charge are concentric circles, centred on the velocity axis of the point charge (see figure).

→ Magnetic field along the line of motion is zero

→ $\frac{1}{\sqrt{\mu_0 \epsilon_0}} = \sqrt{\frac{1}{4\pi\epsilon_0} \cdot \frac{4\pi}{\mu_0}} = \left(\frac{9 \times 10^9}{10^{-7}}\right)^{1/2} = 3 \times 10^8 \, ms^{-1} = c$ thus it must be remembered that $\boxed{1/\sqrt{\mu_0 \epsilon_0} = c}$, this is not just a coincidence, it is rather a consequence of Maxwell's equations of electrodynamics, a rigorous proof of this result will be given in the chapter on electromagnetic waves.

→ Recall that electrostatic field due to a point charge is given as $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q\vec{r}}{r^3}$ (2) Using this result in equation (1), we can

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write $\vec{B} = \mu_0 \epsilon_0 (\vec{v} \times \vec{E}) = (\vec{v} \times \vec{E})/c^2$ thus:

$$\boxed{\vec{B} = (\vec{v} \times \vec{E})/c^2}$$ for $v << c$, here $\vec{E}$ is the electric field produced by point charge at the point where we want to find magnetic field due to it.

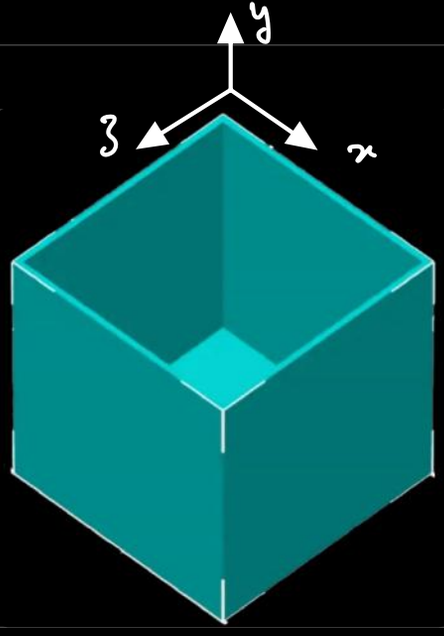
Magnetic Field Due To Charged Translating Rigid Body:

Let a charged body be translating with a velocity $\vec{v}$ in space. Let $\vec{E}_i$ be the electric field produced due to its i^{th} particle at some point in space, then magnetic field produced by it at that point:

$\vec{B}_i = (\vec{v} \times \vec{E}_i)/c^2$, Now summing it up over the entire rigid body:

$$\boxed{\vec{B} = (\vec{v} \times \vec{E})/c^2}$$

→ This result can be conveniently used to find magnetic field due to any translating geometry whose electric field is known.



Example (Original): An open cubical box has a surface charge density σ, find the magnetic field at is body centre if (a) It is moving with a velocity $\vec{v}$ perpendicular to one of the vertical faces (b) It is moving with a velocity $\vec{v}$ perpendicular to base.

Solution: (a) Recall from our discussion on Gauss law for electrostatics, electric field at the centre of this cube is $\vec{E} = \sigma/6\epsilon_0 \hat{j}$.

Now with $\vec{v} \perp \vec{E}$, $\vec{B} = \vec{v} \times \vec{E}/c^2$ thus $B = \sigma v/6\epsilon_0 c^2$ Ans.

(b) Here $\vec{v} \parallel \vec{E}$, thus $\vec{v} \times \vec{E} = 0$, thus $\vec{B} = 0$.

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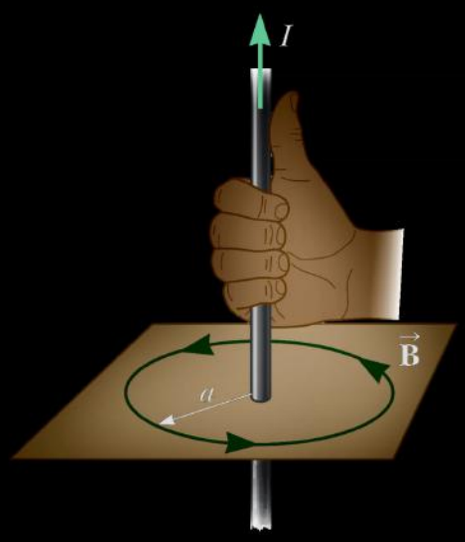
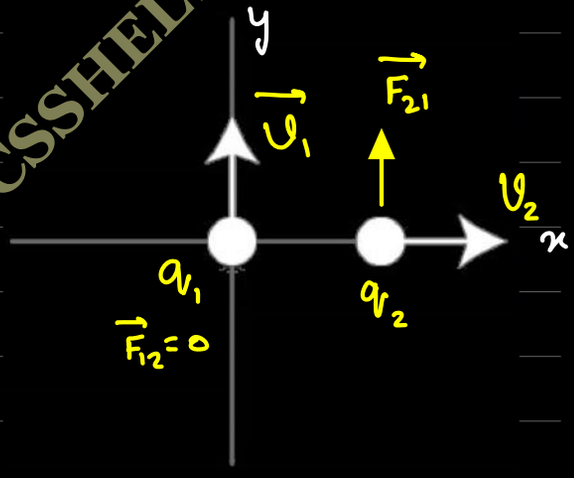
Right Hand Thumb Rule For Visualizing Field Lines:



For quickly visualising field due to a moving charge, a current element on a line current we can quickly find the direction of magnetic field using right hand thumb rule. The rule is as follows: Align the thumb of your right hand along the direction of charge flow and the curl of fingers gives the sense of the surrounding magnetic field lines.

Failure Of Newton's Third Law For Magnetic Forces:



Whenever there are time varying electromagnetic field in space, the Newton's third law as we understand in JEE physics does not hold true. For instance consider two charges moving as shown in the adjacent figure, here we can clearly see that field due to q_2 at q_1 is zero and hence it exerts no magnetic force on q_1 while q_1 exerts a force in $+\hat{j}$ on q_2. Clearly a violation of Newton's third law.

→ Even though standard form of Newton's third law fails here, it turns out that law of conservation of linear momentum still holds if we consider the momentum carried by Electromagnetic field in to account. Whenever there is a time varying electromagnetic field in space, there can be a time varying energy and momentum associated with space. (A photon is nothing but a manifestation of time varying electromagnetic field in space and recall that photon has an energy $h\nu$ and a momentum h/λ. [Also $E = pc$).

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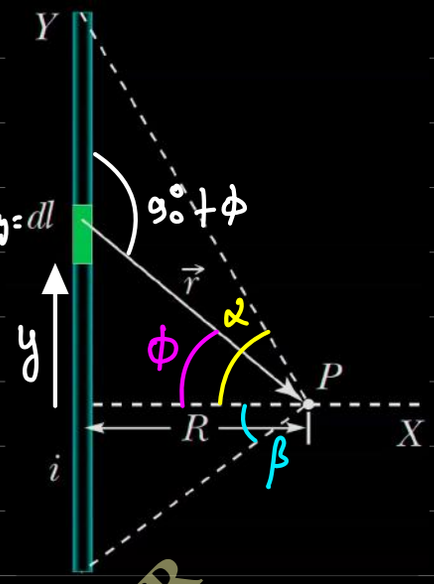
Magnetic Field Due To A Line Segment Current:



Consider an element $i\vec{dl}$ of line segment current as shown in adjacent figure. Now:

$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i\vec{dl} \times \vec{r}}{r^3}$, for our case:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i\,dy \times r \times \sin(90^\circ + \phi)}{r^3} (-\hat{k})$$

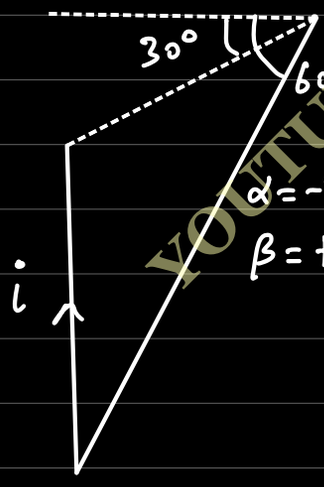
$$\Rightarrow d\vec{B} = \frac{\mu_0}{4\pi} \frac{i}{r^2} \cos\phi\, dy \; (-\hat{k})$$

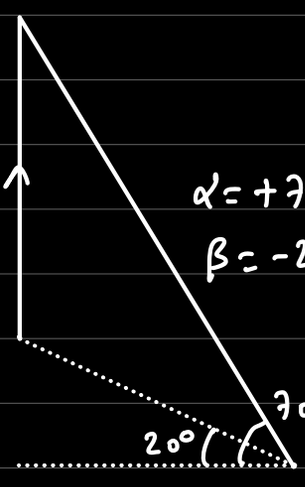
Now $y = R\tan\phi \Rightarrow dy = R\sec^2\phi\, d\phi$, thus,

$$\vec{B} = \frac{\mu_0 i}{4\pi R} \int_{-\beta}^{\alpha} \cos\phi\, d\phi = \frac{\mu_0 i}{4\pi R} [\sin\alpha + \sin\beta] (-\hat{k})$$

Thus field due to line segment $\boxed{B = \frac{\mu_0 i}{4\pi R} [\sin\alpha + \sin\beta]}$

→ Note that α and β are signed values measure from perpendicular to the conductor, α is along the current and β is measured against the current. Thus:





→ If a point lies on the extension of the wire itself then magnetic field there is zero (∵ it is zero individually due to each current element).

→ For infinite wire (i.e wire extends to ∞ both above and below point P) then $\alpha = \beta = 90^\circ$ thus:

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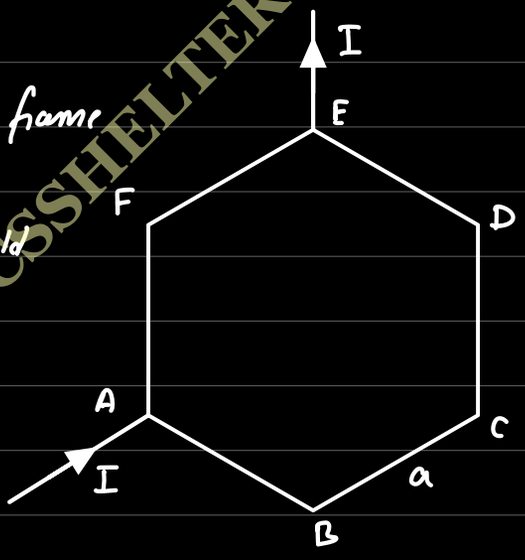
$$B = \frac{\mu_0 I}{2\pi R}$$ ← Field due to infinite wire.

→ For semi-infinite wire (i.e lower end is directly at ⊥ from P and upper extremity is at ∞), $\alpha = 90°$, $\beta = 0°$ thus:

$$B = \frac{\mu_0 I}{4\pi R}$$ ← Field due to semi-infinite wire.

→ Magnetic field due to any current carrying line is always along its skew perpendicular.



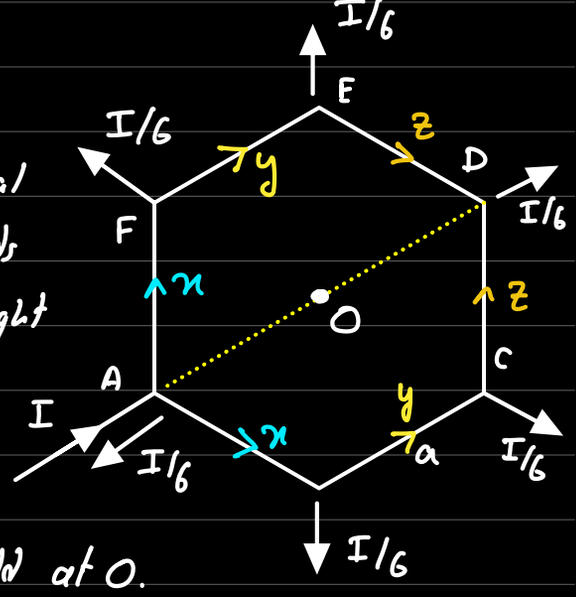
EXAMPLE: A current I enters a hexagonal frame from one of the vertices and leaves from another as shown. Find the magnetic field produced at the centre of the hexagon. Each side of hexagon has a resistance R.

Solution: The current pattern produced can be thought of as superposition of following currents:

(1) Tapping in I from vertex A and tapping out I/6 from all other vertices.

(2) Tapping in I/6 from all vertices and tapping out I from vertex E.

Now considering the first current pattern, we can clearly see that mirror image branches about line of entry carry identical currents and hence produce cancelling fields at O as can be readily inferred from right hand thumb rule, also lines of entry / exits pass through O, thus producing a net zero field at O. Thus overall effect of first pattern is zero magnetic field at O.



Similarly the second pattern would produce a zero field at O.

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Thus superposition of two patterns yields a zero field at O.

→ By same logic, the field at centre would have been zero irrespective of entry or exit vertices, (Just entries and exits should be "radial").

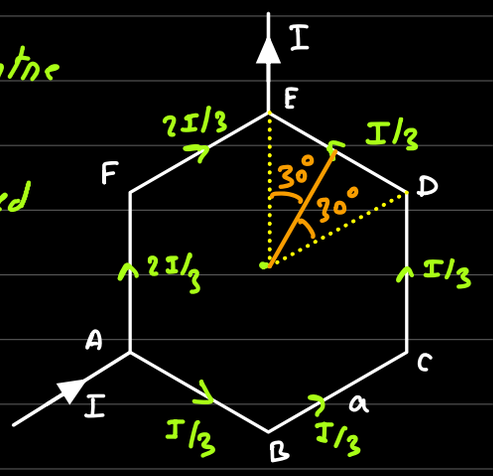
→ By same logic the field would have been zero at the centre even if there were multiple tap in and tap out points because such a pattern would just be a superposition of several single tap in and single tap out point patterns.

→ By same logic the field at O would have been zero even if instead of a hexagon, it were a heptagon, octagon or any other polygon with identical resistances on all its sides.

→ If the resistances were not identical, we could not use the mirror image current logic and thus field at centre would not have been zero in general.

→ A circle is a polygon of infinite sides, thus with radial entry and exits, the field at the centre of a uniform circular wire will be zero. If the entries and exits were not radial, then we will just need to consider fields due to entry and exit wires.

→ We could also find magnetic field at the centre using our formulae for line segment with $\alpha = \beta = 30°$ here (or π/n each for an n sided polygon).



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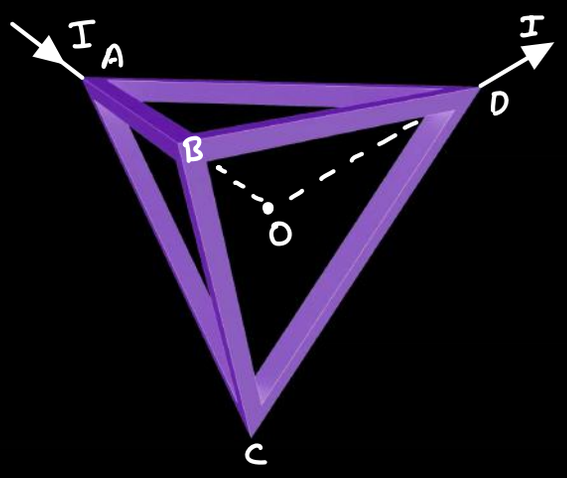
EXAMPLE: 200 MORE PUZZLING: 157:

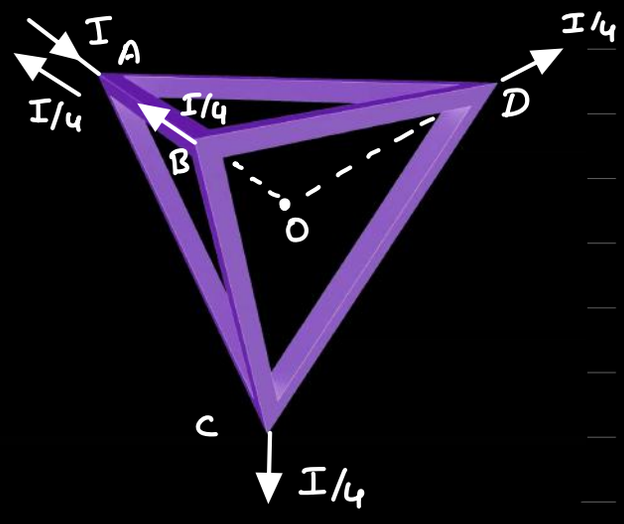
A regular tetrahedron is made of homogeneous resistance wire of uniform cross-section. Current I is conducted into vertex A through a long, straight wire directed towards the centre O of the tetrahedron, and it is conducted away through vertex D in the same way, as illustrated in the figure. What are the magnitude and direction of the magnetic field vector at the centre of the tetrahedron?



Solution: The situation can be thought of as superposition of following two patterns:

(1) Tapping in I from vertex A and tapping out I/4 from each vertex (In this case nothing would flow in branches BD, CD and BC as can be readily seen by junction law)

(2) Tapping in I/4 from each vertex and tapping out I from vertex D.



Here our visualization will become easier by observing the fact that each edge of a regular tetrahedron is skew perpendicular to another edge. For example AB is skew perpendicular to CD, BD is skew perpendicular to AC etc. Now in pattern (1), I/4 current is flowing in each of the branches $\vec{AD}$, $\vec{AB}$ and $\vec{AC}$ causing a magnetic field at centre directed along their respective skew perpendiculars i.e $\vec{BC}$, $\vec{CD}$ and $\vec{DB}$ respectively with same magnitude each. Now since $\vec{BC} + \vec{CD} + \vec{DB} = 0$, therefore due to pattern (1), net field at the centre of tetrahedron must be zero. By same logic field at the centre of tetrahedron due to pattern 2 must also be zero, thus net field at the centre of tetrahedron must be zero. Ans.

→ Not just for tetrahedron, by a careful visualisation, it can be shown that field at centre of all such regular polyhedra (5 platonic polyhedra) is zero irrespective of tap in and tap out points.

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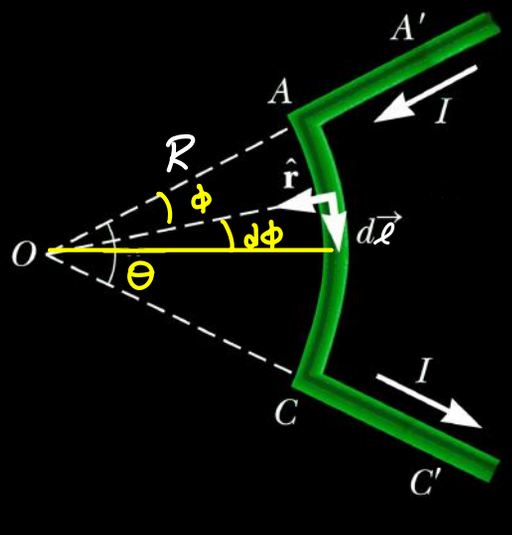
Magnetic Field At Centre Of Curvature Of Current Arc:

Consider an element of angular thickness $d\phi$ at an angular position ϕ. The field due to this element at point O is given as:



$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{i R d\phi \times R \times \sin 90^\circ}{R^3} \quad (-\hat{k})$$

Thus: $$\vec{B} = \frac{\mu_0 i}{4\pi R} \int_0^\theta d\phi \quad (-\hat{k})$$

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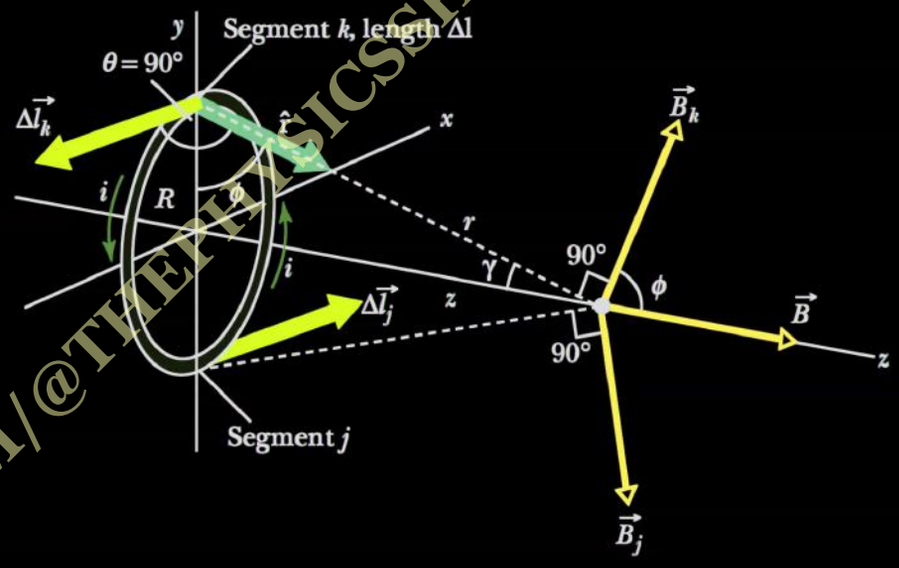
$\Rightarrow$ $\boxed{B = \frac{\mu_0 i \theta}{4\pi R}}$

→ Here the direction of field can also be found using right hand thumb rule for centre of curvature of arcs, i.e curl the fingers of right hand along the arc current and thumb now gives the direction of magnetic field at the centre of curvature.

→ For a complete circular arc, $\theta = 2\pi$ thus for complete arc, $\boxed{B = \frac{\mu_0 i}{2R}}$, thus field due to arc can also be memorized by thinking of it as a $\theta/2\pi$ fraction of a circle i.e $B = \frac{\mu_0 i}{2R} \times \theta/2\pi$.

Magnetic Field On Axis Of A Circular Current Loop:

Consider the adjacent figure. From this figure it can be readily inferred that radial components of magnetic fields produced by 180° opposite elements cancel out thus we only need to consider the axial components of fields for integration. Now consider axial component of field due to element of length $\Delta\vec{l}_k$. Due to this, axial component of field is given as:



$$d\vec{B}_a = \frac{\mu_0}{4\pi} i\, d\vec{l}_k \times \frac{r \times \sin 90° \times \cos\phi}{r^3}$$

Now $r = \sqrt{z^2+R^2}$ and $\cos\phi = R/\sqrt{z^2+R^2}$, thus,

$$d\vec{B}_a = \frac{\mu_0 i}{4\pi} \frac{R}{(z^2+R^2)^{3/2}} \cdot dl_k\, \hat{k}$$

Now taking summation over all elements of the circumference, $\Sigma dl_k = 2\pi R$ thus:

$$\boxed{B = \frac{\mu_0 i}{2} \frac{R^2}{(z^2+R^2)^{3/2}}}$$

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→ If instead of a single turn coil we had an n turn coil (short in length though), then $B = \frac{\mu_0 N i R^2}{2(z^2+R^2)^{3/2}}$

→ Recall that magnetic moment of such a coil is $m = i\pi R^2$, thus in terms of magnetic moment, we can write the magnetic field on the axis as:

$$\boxed{B = \frac{\mu_0}{4\pi} \cdot \frac{2m}{(R^2+z^2)^{3/2}}}$$

→ For $z >> R$, the loop can be considered as a point magnetic dipole with magnetic field given as:

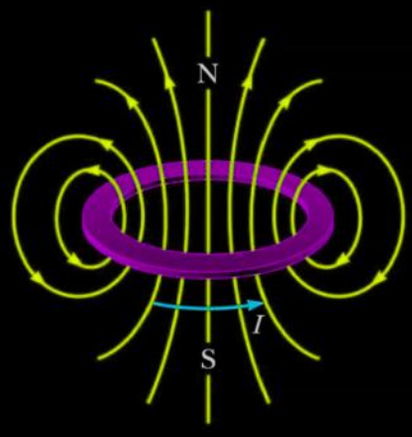
$$\boxed{B_a = \frac{\mu_0}{4\pi} \cdot \frac{2m}{z^3}} \quad \text{or} \quad \boxed{\vec{B}_a = \frac{\mu_0}{4\pi} \frac{2\vec{m}}{r^3}}$$

This is just like the electric field of an electric dipole. A point current dipole is thus very similar to point electric dipole for purposes of field calculation, potential energy and torque calculations and force calculations.

→ For finding direction of axial field, once again we can use the right hand thumb rule i.e curl the fingers of right hand along the current and thumb gives the direction of field.

→ In case of a point current loop (or magnetic dipole), the field in the equatorial position is also similar to that of an electric dipole thus:

$$\boxed{\vec{B}_E = -\frac{\mu_0}{4\pi} \frac{\vec{m}}{r^3}}$$



→ Thus for same distance, field in equatorial position is half the field on the axial position.

→ Consider the formulae: $\vec{\tau} = \vec{m} \times \vec{B}$, $U = -\vec{m} \cdot \vec{B}$, $\vec{F} = (\vec{m} \cdot \nabla)\vec{B}$ $\vec{B}_a = k_m \frac{2\vec{m}}{r^3}$ and $\vec{B}_E = -k_m \frac{\vec{m}}{r^3}$ where $k_m = \mu_0/4\pi$,

all these formulae are exact parallels for the ones that we obtained for electric dipoles. Thus magnetic field is to a point loop

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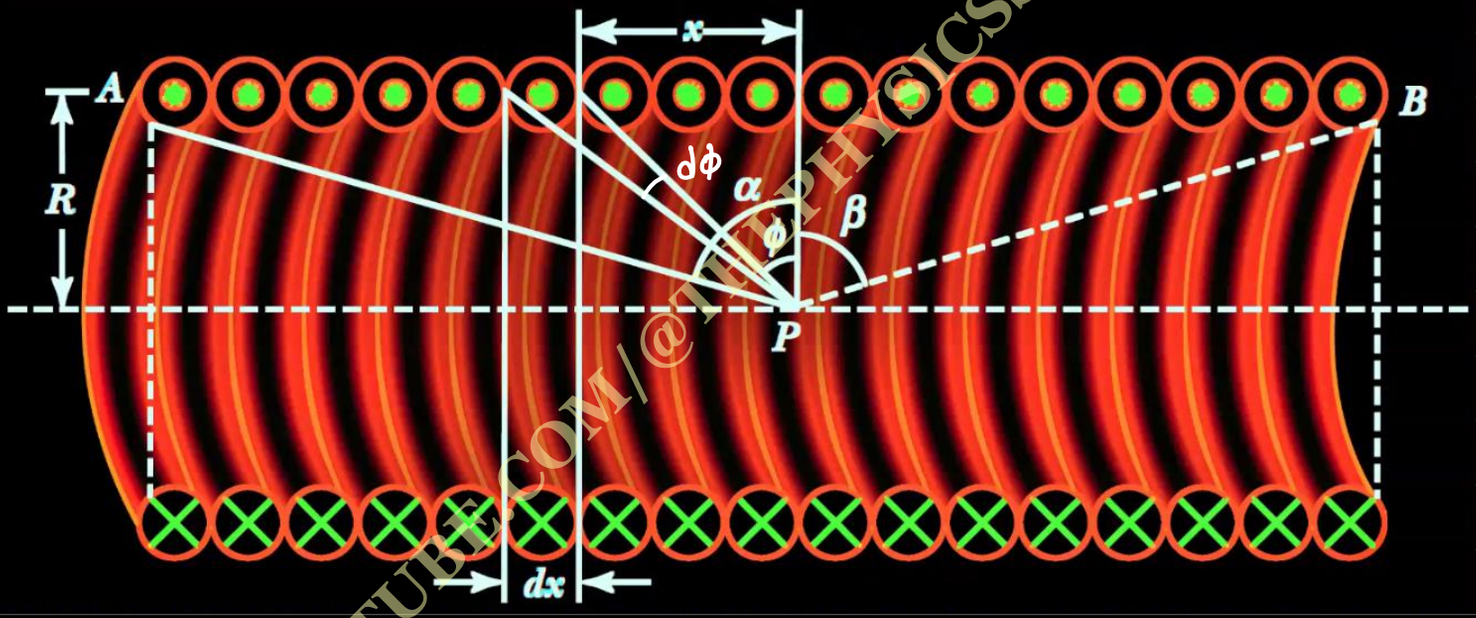
what electric field is to a point dipole. The only difference here is that magnetic field lines form perfectly closed loops (magnetic monopoles do not exist), while electric monopoles do exist.

Magnetic Field Due To A Solenoid:

A solenoid is simply a copper coil wound over a cylindrical core. The solenoids may have an air core or an iron core depending on the application purpose. For our purpose here we will be dealing with an air core solenoid which can be obtained by winding a copper wire over a hollow plastic cylinder (or any other material hollow cylinder).



For purpose of finding the field, a solenoid can be modeled as a stack of several coaxial current carrying rings. Let n be the number of turns per unit length. We wish to find the field at a point P on the axis of the solenoid. Let radius of the solenoid be R and angles subtended by point P at extremities be α and β as shown.

Now consider an element of solenoid of thickness dx at a position x from point P, further let the angular position and angular thickness of this element be ϕ and $d\phi$ respectively. Now recall that field on axis of a ring is given as $B = \frac{\mu_0}{2} \frac{iR^2}{(z^2+R^2)^{3/2}}$, for our element, number of turns is ndx. Thus elemental field is

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given as:

$$dB = \frac{\mu_0}{2} \frac{i R^2 n dx}{(x^2+R^2)^{3/2}}, \quad \text{Also } x = R\tan\phi$$

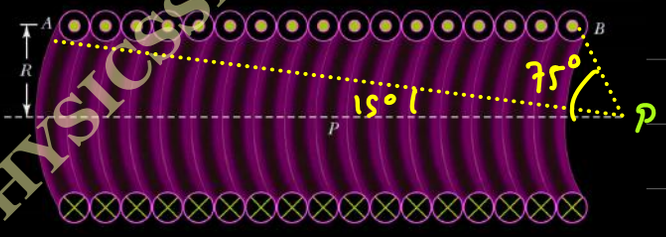
Thus $dx = R\sec^2\phi\, d\phi$, Thus our elemental field becomes:

$$\frac{\mu_0}{2} \frac{i R^2 n \times R\sec^2\phi\, d\phi}{(R^2\tan^2\phi + R^2)^{3/2}} = \frac{\mu_0 n i}{2} \cos\phi\, d\phi, \quad \text{Now Integrating:}$$

$$B = \frac{\mu_0 n i}{2} \int_{-\beta}^{\alpha} \cos\phi\, d\phi = \frac{\mu_0 n i}{2} [\sin\alpha + \sin\beta], \text{ thus:}$$

$$\boxed{B = \frac{\mu_0 n i}{2} [\sin\alpha + \sin\beta]}$$

→ Note that α and β are signed values, for example in the adjacent figure, for finding field at P, $\alpha = 75°$ and $\beta = -15°$



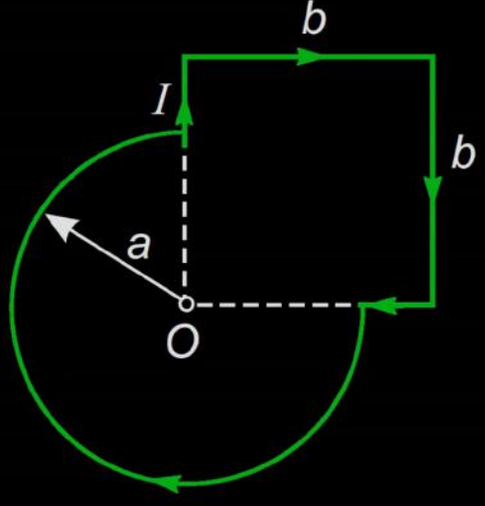
→ For a point P well within a very long (modeled as ∞) solenoid, $\alpha = \beta = 90°$, thus for very long solenoid, field at axis:

$$\boxed{B = \mu_0 n i}$$

→ For point at the centre of end face of a long solenoid, $\alpha = 90°$, $\beta = 0°$ thus $B = \mu_0 n i/2$.

EXAMPLE: IRODOV: 3.223C:

Find the magnetic field due to shown current loop at point O.



Solution: The given loop can be modeled as a combination of arcs and line segments. Recall that field due to arc: $B = \frac{\mu_0 i}{2R} \times (\theta/2\pi)$ and due to line segment,

$$B = \frac{\mu_0 i}{4\pi R} (\sin\alpha + \sin\beta), \text{ Using these}$$

formulae, B at O in our case:

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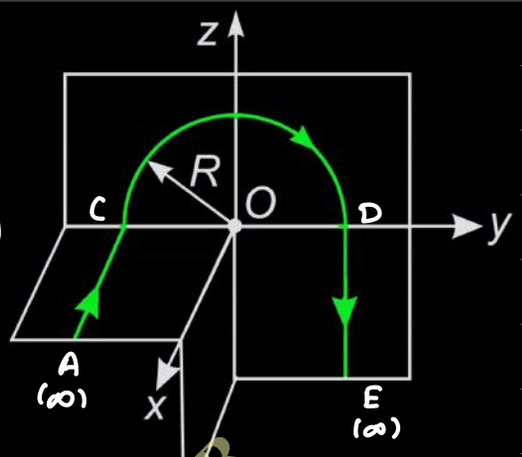
$$\frac{\mu_0 i}{2a} \times \frac{3\pi/2}{2\pi} + \frac{\mu_0 i}{4\pi b}\left[\sin 0° + \sin 45°\right] \times 2 = \frac{\mu_0 i}{4\pi}\left[\frac{3\pi}{2a} + \frac{\sqrt{2}}{b}\right]$$ Ans.

EXAMPLE: IRODOV: 3·228b: Find the magnetic field at point O if the wire carries a current $i = 8A$. The radius of curved part is $R = 100mm$ and straight sections are infinite.



Solution: Field due to section AC: $\vec{B}_1 = \frac{\mu_0 i}{4\pi R}(-\hat{k})$

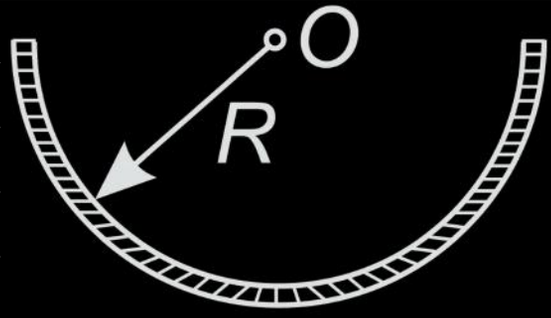
Due to $\overset{\frown}{CD}$ $B_2 = (\mu_0 i/2R)(\pi/2\pi)(-\hat{i})$

$\Rightarrow \vec{B}_2 = \frac{\mu_0 i}{4R}(-\hat{i})$ (Use thumb rule).

Due to DE: $\vec{B}_3 = (\mu_0 i/4\pi R)(-\hat{i})$

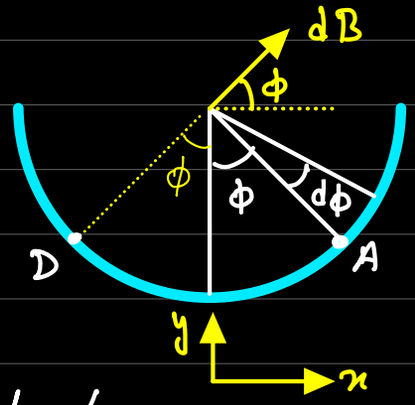
Now net field, $\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = -\frac{\mu_0 i}{4\pi R}\left[(\pi+1)\hat{i} + \hat{k}\right]$ Ans.

EXAMPLE: IRODOV: 3·225: A current I flows in a long straight wire having a cross section in form of a thin semi-circular ring. Find the magnetic field at point O on the axis (far away from ends).



Solution: Consider an element of the conductor with angular thickness $d\phi$ at an angular position ϕ as shown. This is like a thin straight wire, due to this element,

$$dB = \frac{\mu_0}{2\pi R}\left(\frac{i}{\pi}\right)d\phi$$



Now by symmetry, x components of fields due to elements add up, while y components cancel out (for instance field due to element near D will be ϕ below horizontal)

Now Integrating x component of the field:

$$B = \frac{\mu_0 i}{2\pi^2 R}\int_{-\pi/2}^{\pi/2} \cos\phi \, d\phi = \frac{\mu_0 i}{\pi^2 R}$$ Ans.

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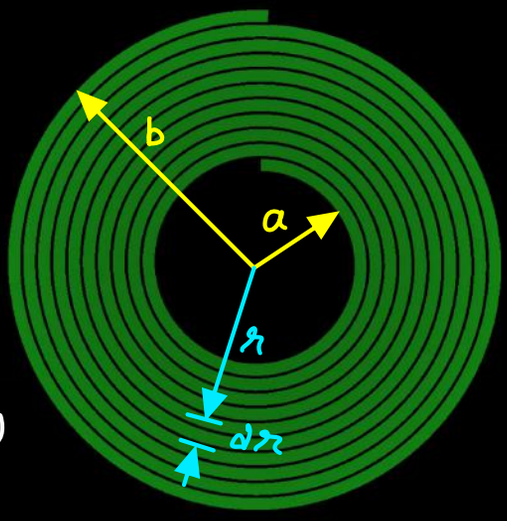
Example: Irodov: 3.245: A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8\,mA$. The radii of inside and outside turns are equal to $a = 50\,mm$ and $b = 100\,mm$. Find the magnetic induction at the centre of the coil.

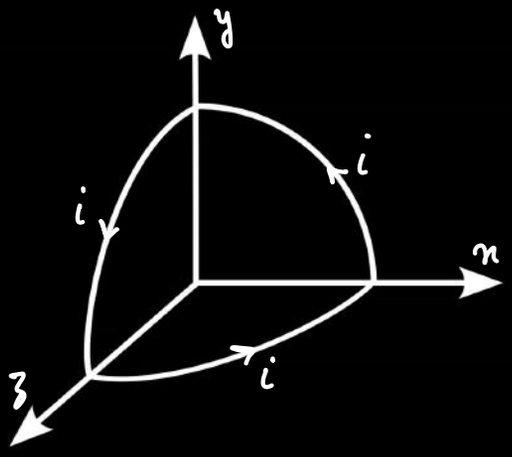


Solution: No of turns / unit thickness: $n = N/(b-a)$

Now consider an element of the coil at radius r and thickness dr. The magnetic field at centre of the coil due to this element: $dB = \frac{\mu_0 I}{2r} \times \frac{N}{(b-a)}\,dr$. Now Integrating:

$$B = \frac{\mu_0 N I}{2(b-a)} \int_a^b \frac{1}{r}\,dr = \frac{\mu_0 N I}{2(b-a)} \ln(b/a)$$ Ans. (Value: $7\,\mu T$).

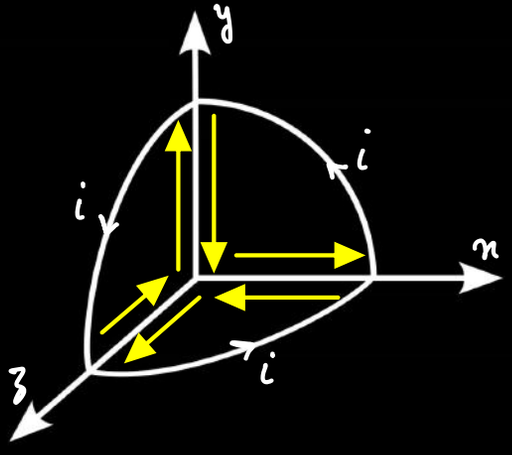
Example: Consider the three dimensional current carrying loop made from three quarter circular arcs each having a radius R. (a) What is the magnetic moment of the loop (b) What is the magnetic field due to the loop at a point $\vec{r} = r(\hat{i}+\hat{j}+\hat{k})/\sqrt{3}$, $r >> R$.



Solution: We can break the loop in to three planar loops by imagining cancelling currents along x, y and z axes as shown in the figure. Thus, $\vec{m} = (\pi R^2/4)[\hat{i}+\hat{j}+\hat{k}]\,i$

Now recognize that given $\vec{r}$ is a point along axial position, thus:

$$\vec{B} = \frac{2k_m \vec{m}}{r^3} = \frac{\mu_0 i}{2\pi} \cdot \left(\frac{\pi R^2}{4}\right) \frac{(\hat{i}+\hat{j}+\hat{k})}{r^3}$$



Thus $\vec{B} = \frac{\mu_0 i}{8} \frac{R^2}{r^3}$ Ans.

→ Just like an electric dipole, for point magnetic dipole, field at arbitrary position vector is $\vec{B} = \frac{k_m}{r^3}(3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m})$.

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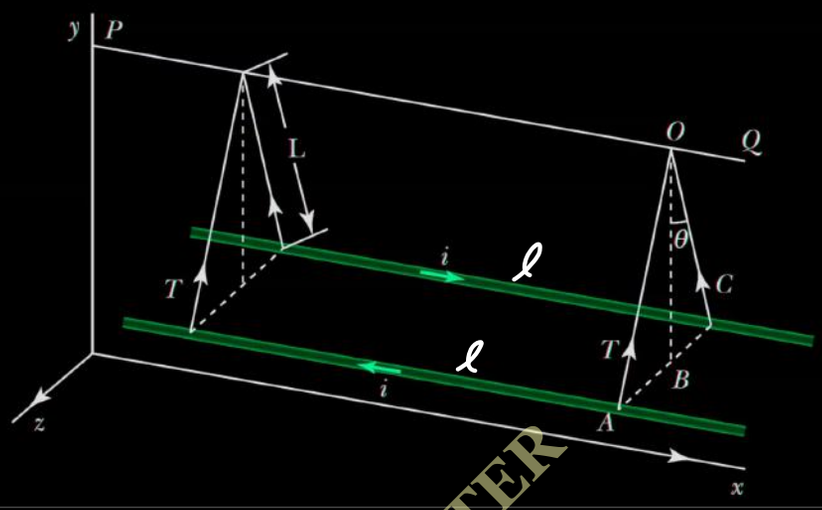
Where $k_m = \mu_0/4\pi$

EXAMPLE :

Two long parallel wires carry currents i of equal magnitudes but in opposite directions. These wires are suspended from rod PQ by four chords of same length L as shown in Fig. The mass per unit length of wires is λ. Determine the value of θ at equilibrium (assuming it to be small).



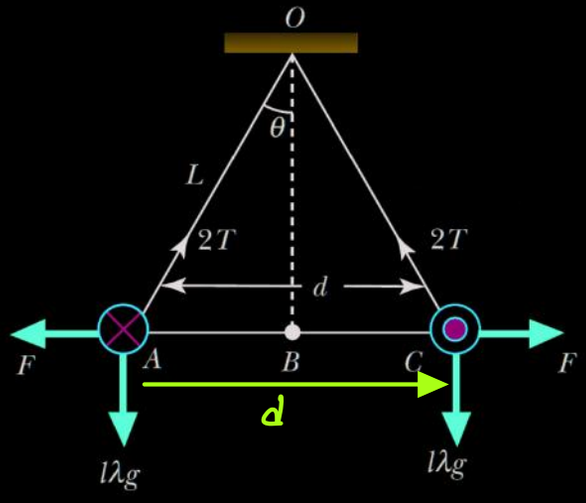
Solution: Field of one wire at the other, $B = \mu_0 i/2\pi a = \mu_0 i/4\pi L\theta$ ($\because \theta$ is small). Thus force of repulsion between wires, $\vec{F} = i\vec{l} \times \vec{B} \Rightarrow$

$F = \mu_0 i^2 l/4\pi L\theta$. Now balancing the forces on one wire:

$2T\cos\theta = \lambda l g$ and $2T\sin\theta = \mu_0 i^2 l/4\pi L\theta$, Dividing we get:

$\tan\theta = \dfrac{\mu_0 i^2}{4\pi\lambda L g\theta}$, but for small θ, $\tan\theta \approx \theta$, thus:

$$\theta^2 = \frac{\mu_0 i^2}{4\pi\lambda L g} \Rightarrow \theta = i\sqrt{\frac{\mu_0}{4\pi\lambda L g}}$$



→ Unlike currents repel each other

→ One Ampere is defined as current which must be flown through parallel infinite wires separated by 1m so as to produce in them a magnetic force of interaction equal to 2×10^{-7} N/m. This can be shown as follows: $B = \dfrac{\mu_0 i}{2\pi d} \Rightarrow F = \dfrac{\mu_0 i^2 l}{2\pi d}$ ($\because F = ilB$) $\Rightarrow F/l = \dfrac{\mu_0 i^2}{2\pi d}$

Now when $i = 1$, $d = 1 \Rightarrow \dfrac{F}{l} = 2\times10^{-7}$ N/m. This is the classical definition of the unit 1A.

DIRECTED LOOP: A directed loop is just any closed path to which we have assigned the direction of traversal.

The idea of directed loop is required for stating and understanding Ampere's law which we shall learn shortly.

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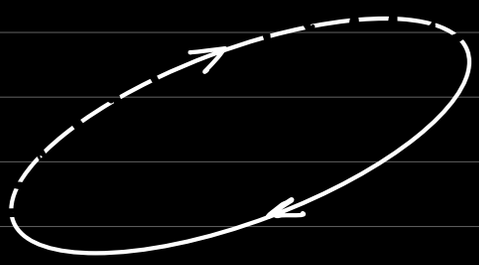
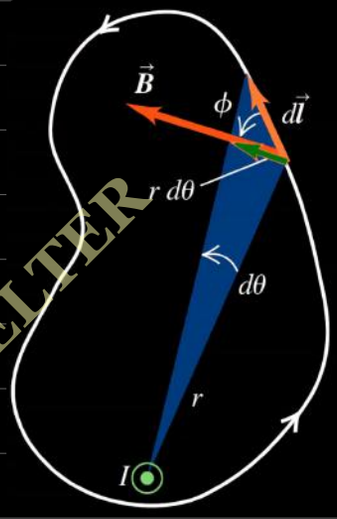
Once the sense of the directed loop is decided, we can uniquely define its area normal using the right hand thumb rule, i.e to find the direction of area normal, curl the fingers of your right hand along the sense of the loop now thumb gives the definition of area normal vector.



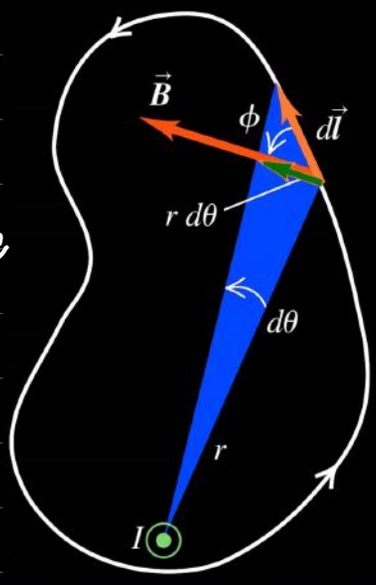
Circulation Of A Vector: Let $\vec{B}$ be a vector field defined in space then we define the circulation of $\vec{B}$ over any directed loop as: $\oint \vec{B} \cdot d\vec{l}$

where $d\vec{l}$ is elemental segment of the directed loop and the circle over the integration sign shows that the integration is to be done along the entire closed path.



Circulation Of Magnetic Field Due To An Infinite Wire:

Consider a loop around a current carrying wire such that current flows along directed area normal. Now if we consider the small element $d\vec{l}$ of the wire located at a position $\vec{r}$ wrt wire as shown, $d\vec{l} = r\,d\theta\,\hat{\theta} + dr\,\hat{r}$, Now $\vec{B} = \frac{\mu_0 i}{2\pi r}\hat{\theta}$, thus,



$\vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2\pi} d\theta$, note that this term only depends on the angular width of element subtended on the wire and not its absolute distance.

Now integrating the elemental circulation, $\oint \vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2\pi} \times \int d\theta$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

→ For any loop portion, subtending net angle θ on wire, $\int \vec{B} \cdot d\vec{l} = \frac{\mu_0 i \theta}{2\pi}$.

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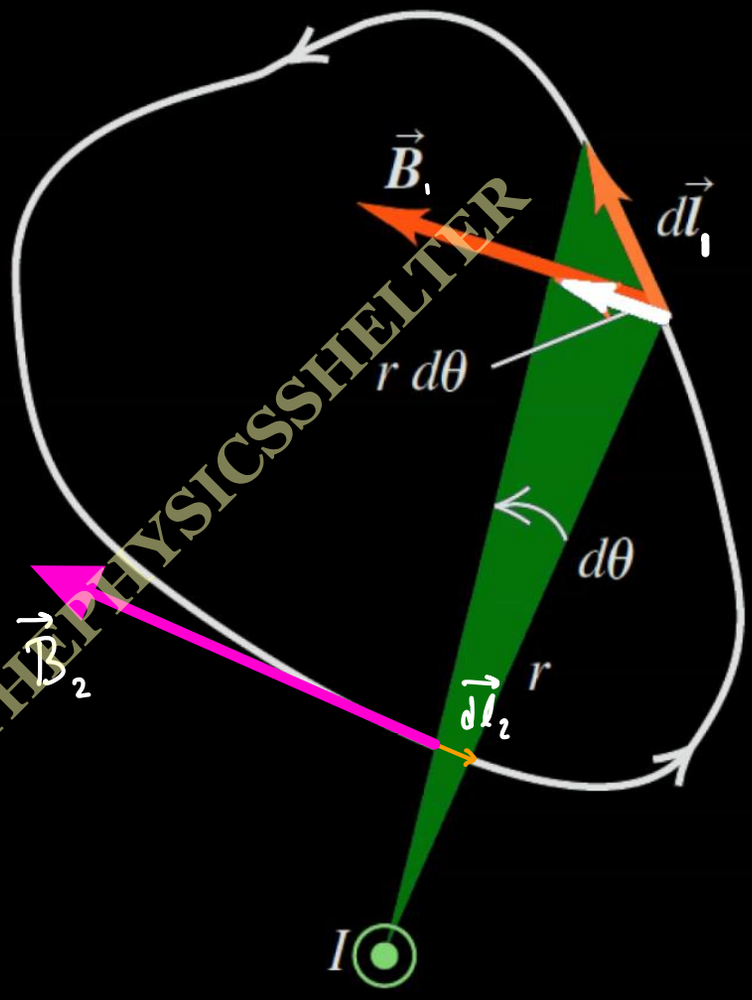
→ For a 3D loop, $d\vec{l} = r\,d\theta\,\hat{\theta} + dr\,\hat{r} + dz\,\hat{z}$, however on taking $\vec{B}\cdot d\vec{l}$, $\hat{r}$ and $\hat{z}$ terms will go to zero, thus $\vec{B}\cdot d\vec{l}$ only depends on tangential projection of $d\vec{l}$, thus even if we had a 3D loop around I, $\oint \vec{B}\cdot d\vec{e} = \mu_0 i$

Circulation Of Magnetic Field Due To A Wire Outside The Loop:

Recall that in the last section, we proved that circulation of magnetic field of an infinite wire associated with a $d\vec{l}$ just depends on angular projection of $d\vec{l}$ and is simply $\frac{\mu_0 i}{2\pi} d\theta$.



Now consider a wire which is outside the loop. We consider two segments $d\vec{l}_1$ and $d\vec{l}_2$ of the loop lying in the same sector of angular width $d\theta$. (see figure). We can clearly see that for $d\vec{l}_1$, $\vec{B}$ makes an acute angle, while $d\vec{l}_2$, $\vec{B}$ makes an obtuse angle, thus even though $\vec{B}\cdot d\vec{l}$ has same magnitudes for $\vec{B}\cdot d\vec{l}_1$ and $\vec{B}\cdot d\vec{l}_2$, but the signs are just the opposite thus $\vec{B}\cdot d\vec{l}_1 + \vec{B}\cdot d\vec{l}_2 = 0$. In the same way we can see pairwise cancellations of $\vec{B}\cdot d\vec{l}$ in every sector of the loop thus now for the loop as a whole, $\oint \vec{B}\cdot d\vec{l} = 0$.

Calculating Current Enclosed By A Loop: Just like we stated Gauss' law in electrostatics which was related to charges enclosed in the Gaussian surfaces, similarly we shall be stating the Ampere's law which is related to currents enclosed by the "Amperian" loops or directed loops. For finding the sign of any current crossing a loop, just see if the direction of crossing is making an acute angle

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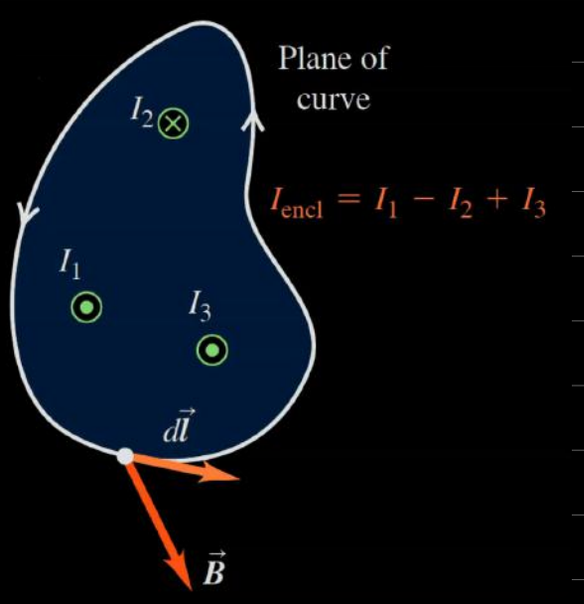
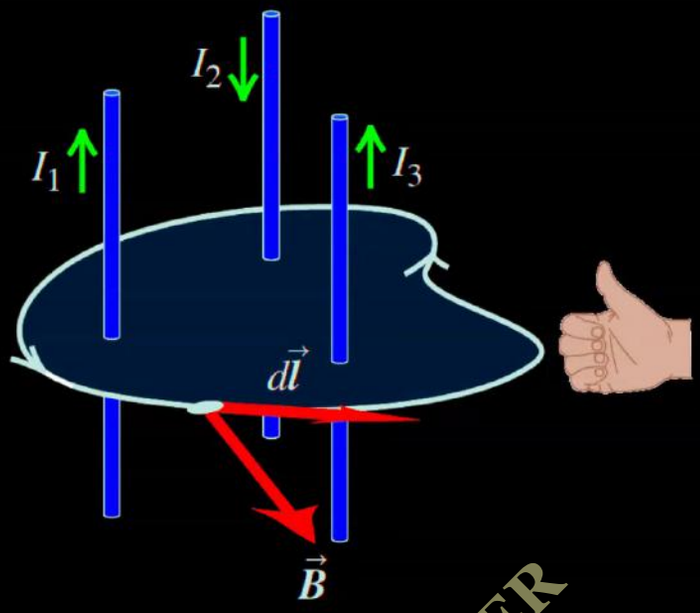


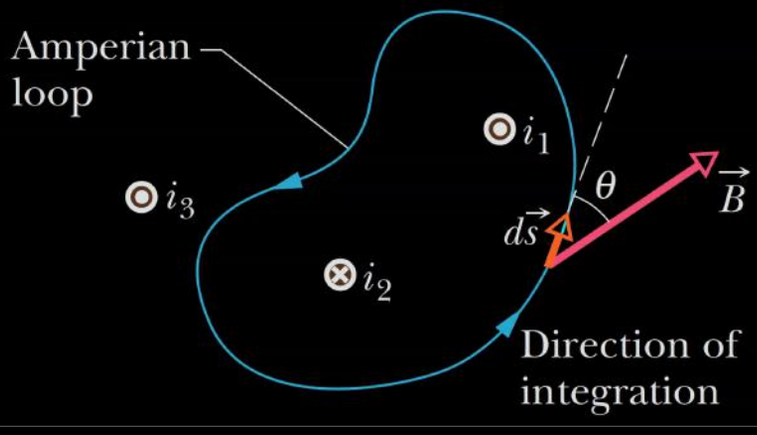


with the directed area normal or an obtuse angle. If the direction of crossing makes an acute angle with directed Area normal, the enclosed current is considered as positive, if it makes obtuse angle with Area Vector, it is considered as negative. (Directed Area Normal is found by curling the fingers of the right hand in the sense of the loop and then looking at the direction in which the thumb points).

Ampere's Law:

The net circulation of magnetic field along any directed loop is equal to μ_0 times the current enclosed by the loop or Mathematically:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{enc}$$



→ For applying Ampere's law, sign of current must be carefully considered. For instance for the shown loop, $\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_1 - i_2)$.

→ Currents outside a loop do not contribute to net circulation along a loop.

→ Even though we have proved Ampere's law with infinite conductors, it also holds true for any arbitrary current loop. (Loop because for Ampere's law in this form, there should be no time varying

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Electrical flux through the Amperian loop).

→ In presence of time varying electric fields, Ampere's law gets modified to Ampere–Maxwell law i.e. $\oint \vec{B}\cdot d\vec{l} = \mu_0\left(i_{enc} + \epsilon_0 \frac{d\Phi_E}{dt}\right)$

where Φ_E represents electric flux through the Amperian loop (Amperian surface). We shall explore Ampere Maxwell law more in the chapter on Electromagnetic waves.

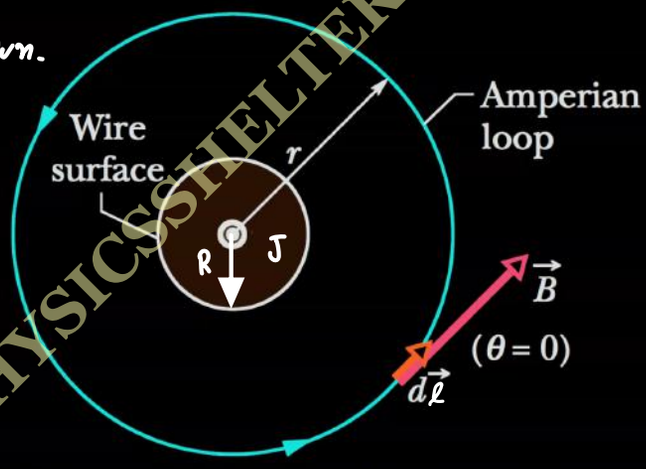
Field Due To A Cylindrical Conductor With Uniform Current Density:

Consider a cylindrical wire carrying a current density J coming out of the page as shown. Now let us consider an Amperian loop of radius r ($>R$). Applying Ampere's law:

$$B \times 2\pi r = \mu_0 \times J \times \pi R^2$$

$$\Rightarrow \boxed{B = \mu_0 J R^2/2r}$$

Vectorially, $\boxed{\vec{B} = \frac{\mu_0 R^2 \vec{J}\times\vec{r}}{2r^2}}$

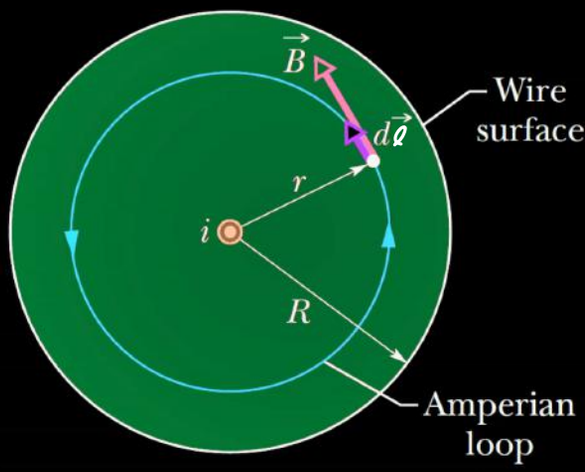


Now for $r < R$,

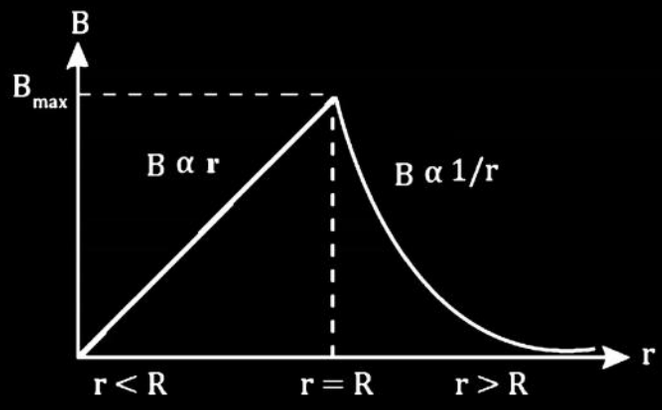
$$B \times 2\pi r = \mu_0 J \pi r^2$$

$$\Rightarrow \boxed{B = \mu_0 J r/2}$$

Vectorially, $\boxed{\vec{B} = \mu_0 \frac{\vec{J}\times\vec{r}}{2}}$



→ B vs r graph for this current geometry is shown in adjacent figure. Note that inside the cylinder, the magnetic field increases linearly and outside the cylinder, it declines hyperbolically. The maximum field occurs at the surface.



→ The electric fields due to similar geometry were $\rho R^2/2\epsilon_0 r$ and $\rho r/2\epsilon_0$ respectively for outside

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and inside points respectively where ρ is the volumetric charge density of the cylinder.

→ Recall that for translating charges $\vec{B} = \vec{v} \times \vec{E}/c^2$ we could use this formula also for calculating the magnetic field. For our case, $J = \rho \pi R^2 v / \pi R^2$ i.e ρv ($\because dq = \rho \times \pi R^2 \times v dt$).

Now for outside point $E = \rho R^2/2\epsilon_0 r \Rightarrow B = \dfrac{\rho R^2 v}{2\epsilon_0 r c^2}$, $c = \dfrac{1}{\sqrt{\mu_0 \epsilon_0}}$

$\Rightarrow B = \mu_0 J R^2/2r$ QED. Similarly for inside point, $E = \rho r/2\epsilon_0$

Thus $B = \dfrac{v \times E}{c^2} = \dfrac{\rho v r}{2\epsilon_0 c^2} = \mu_0 J r/2$ QED.

EXAMPLE: IRODOV 3.234.

Inside a long straight uniform wire of round cross-section there is a long round cylindrical cavity whose axis is parallel to the axis of the wire and displaced from the latter by a distance l. A direct current of density j flows along the wire. Find the magnetic induction inside the cavity.

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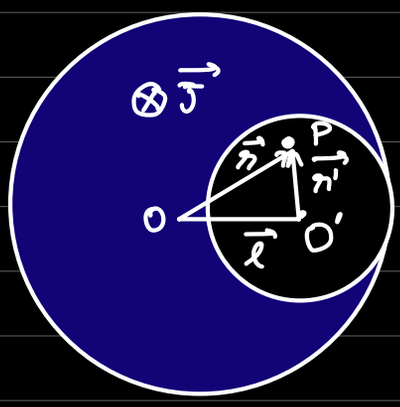
Solution: Method 1: The portion with cavity can be modeled as superposition of $\vec{J}$ and $-\vec{J}$. Now consider an arbitrary point P in the cavity with position vector $\vec{r}$ and $\vec{r'}$ wrt the centres O and O' of the cylindrical regions. Now by superposition, field:



$$\vec{B} = \frac{\mu_0 \vec{J} \times \vec{r}}{2} - \frac{\mu_0 \vec{J} \times \vec{r'}}{2} = \mu_0 \vec{J} \times (\vec{r} - \vec{r'})/2 \Rightarrow \vec{B} = \frac{\mu_0 \vec{J} \times \vec{l}}{2}$$ Ans.

Method 2: Let ρ be the charge density of moving charges, then $J = \rho v$

Now recall that E in such a cylindrical cavity came out to be $E = \rho l/2\epsilon_0$, thus $B = |\vec{v} \times \vec{E}/c^2| = \dfrac{\rho l}{2\epsilon_0} \times \dfrac{v}{c^2} = \dfrac{\mu_0 J l}{2}$ Ans. QED.

Think: Can you infer "shell theorems" for cylindrically symmetric axial flows i.e. for outside points, it behaves like thin current wire and inside it produces zero field.

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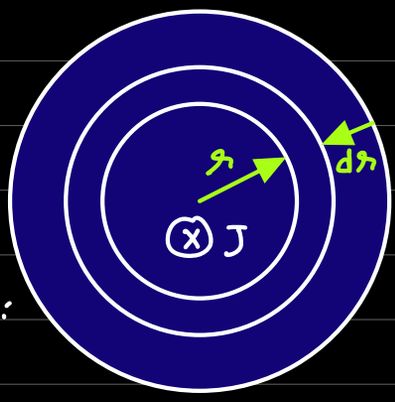
Cylindrically Symmetric Axial Current Density:



Let I be current enclosed up to radius r and let $I + dI$ be the current enclosed up to radius $r + dr$, further let B and $B + dB$ be the magnetic fields at radii r and $r + dr$ then:

$$B \times 2\pi r = \mu_0 I, \text{ Now let us move a distance } dr:$$

$$\Rightarrow 2\pi [B\,dr + r\,dB] = \mu_0 dI = \mu_0 J \times 2\pi r\,dr$$

Now rearranging the equation:

$$\boxed{\frac{dB}{dr} + \frac{B}{r} = \mu_0 J}$$

, Alternatively we could also derive this result using analogous equation in electrostatics.

Recall that $\frac{dE}{dr} + \frac{E}{r} = \rho/\epsilon_0$, Multiplying entire equation by v/c^2:

$$\frac{d}{dr}\left(\frac{Ev}{c^2}\right) + \frac{Ev}{rc^2} = \frac{\rho v}{\epsilon_0 c^2} \Rightarrow \frac{dB}{dr} + \frac{B}{r} = \mu_0 J \quad \text{QED.}$$

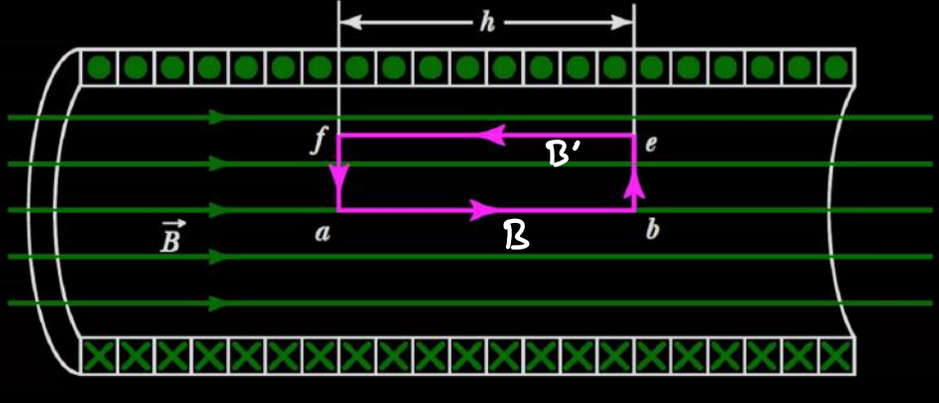
Example: Inodov 3.235:

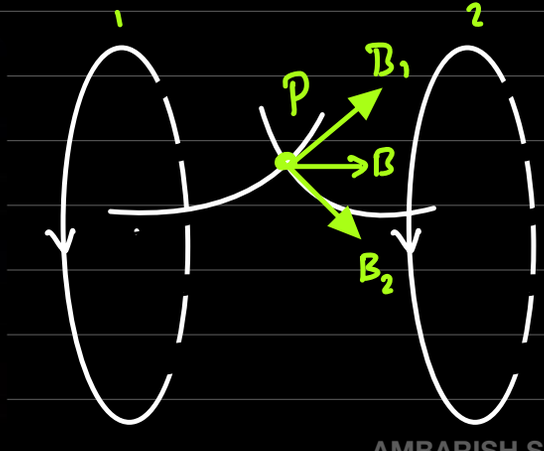
Find the current density as a function of distance r from the axis of a radially symmetrical parallel stream of electrons if the magnetic induction inside the stream varies as $B = br^{\alpha}$ where b and α are positive constants.

Solution: $dB/dr + B/r = \mu_0 J = b[\alpha r^{\alpha-1} + r^{\alpha-1}] = \mu_0 J$

$$\Rightarrow J = b(\alpha+1) r^{\alpha-1}/\mu_0 \quad \text{Ans.}$$

Field Due To An Infinite Solenoid:





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Inside an infinite solenoid for every point P inside the solenoid one can always find equidistant loops from P which produce a net field parallel to solenoid axis, thus field everywhere inside the infinite solenoid must be parallel to the axis of the solenoid. Recall that earlier we had proved that field on the axis of a long solenoid is $\mu_0 ni$, now using Ampere's law we will prove that not only on the axis but everywhere inside an infinite solenoid, the field is uniform.

Consider an Amperian loop abefa inside the solenoid having a width h.

Now applying Ampere's law: $Bh - B'h = \mu_0 \times 0 \Rightarrow B' = B$.

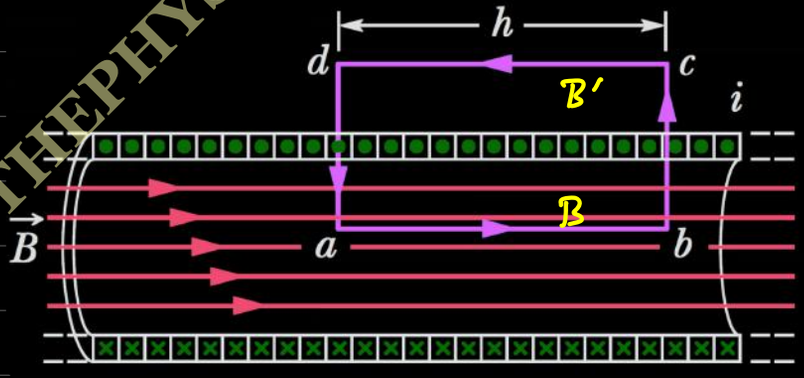
Here B' is the field at off-axis point and B is field at the axis point. Thus we proved that field at an off axis point is same as that at the axis. Thus field everywhere inside an infinite solenoid (Away from ends) is simply:

$$\boxed{B = \mu_0 ni}$$

Field At Outside Points:

Now consider a rectangular Amperian loop which is partially outside the solenoid. We showed that if at all there is field outside the solenoid it, must be parallel to axis of solenoid. Let this field be B'. Now applying Ampere's law: $Bh - B'h = \mu_0 nhi \Rightarrow \mu_0 nih - B'h = \mu_0 nih$

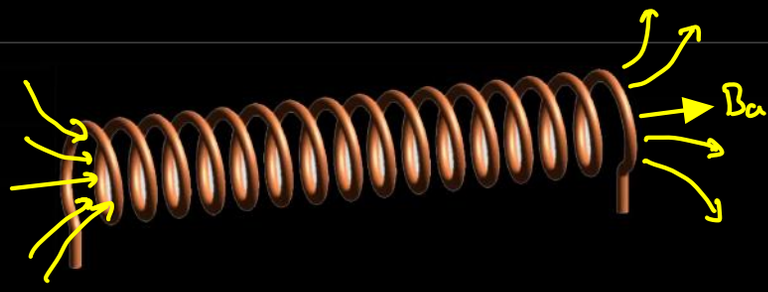
$\Rightarrow B' = 0$. Thus everywhere outside the infinite solenoid, the field is zero.



Axial Field At The Endface Of Infinite Solenoid:

Let B_a be the axial field at the end face of the solenoid. Now if two such solenoids were joined end to end, each point of end face will become a point well within the joint solenoid. Thus $2B_a = B$ or $B_a = B/2 = \frac{\mu_0 ni}{2}$

Here B is field at a point well within the solenoid.



$$\boxed{B_a = \mu_0 ni/2}$$

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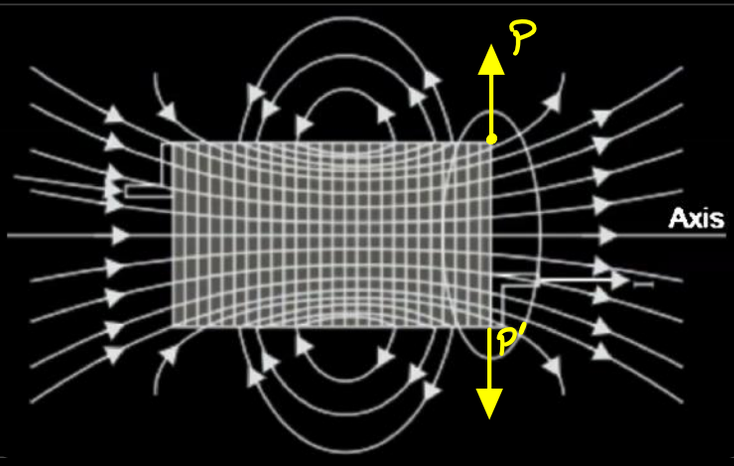
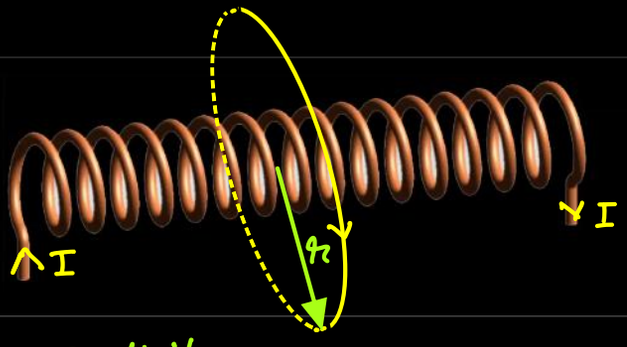
Field Lines Of A Real Solenoid:



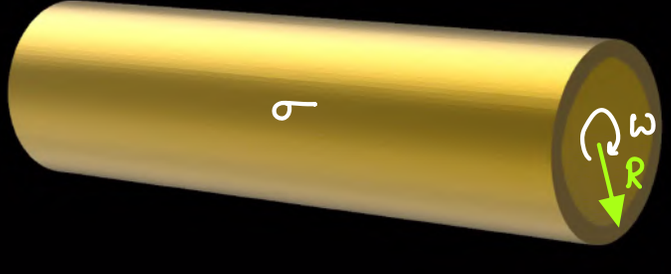
In case of a real solenoid, the field line pattern looks as shown in adjacent figure. If the solenoid is fairly long, the field remains equal to $\mu_0 ni$ for a fairly large region, the field lines remain nearly parallel well within the solenoid, however near the ends, field lines begin to flare out and start emerging from the curved surface. Only 50% of the field lines are able to reach the end faces (recall that $B_a = B/2$ thus flux from end section will be exactly half the flux at a point well within an infinite solenoid).

→ Field line just outside point P (end face) of an infinite solenoid must be $\perp$ to the solenoid axis because if a similar solenoid were to be joined end to end, it must result in a zero field outside P. Thus field lines at P must be cancelling after joining, thus by symmetry, field line just outside P must be $\perp$ to solenoid axis.

→ So far we assumed solenoids to be made of perfectly circular turns, while in practice the turns are helical thus causing an overall axial current as well, this creates a tangential field whose one of the field lines is shown. $B \times 2\pi r = \mu_0 i \Rightarrow B = \mu_0 i/2\pi r$. Thus tangential field is just like that of a line of current.



→ Sometimes instead of thinking $B = \mu_0 ni$ it is useful to think as $B = \mu_0 \lambda$ where λ is current per unit axial length. ($\because n$ is turns per unit axial length). For instance consider a thin cylindrical shell of radius R having charge density σ and rotating with ω, the λ for this is: $\lambda = \sigma \times 2\pi R \times \omega/2\pi$ (i.e charge/unit length $\times v$).



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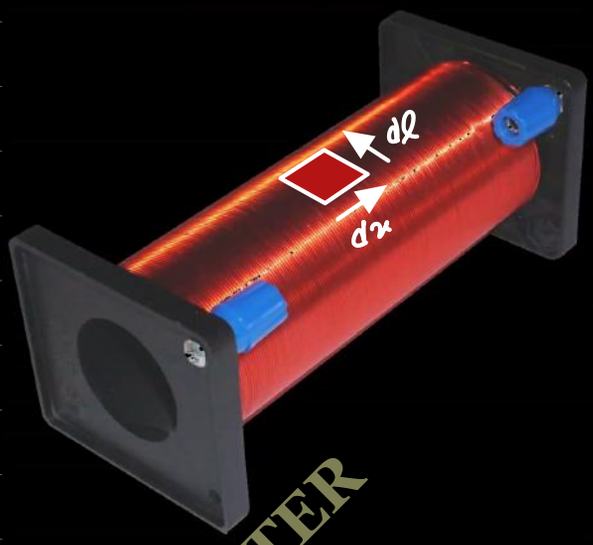
Thus $\lambda = \sigma\omega R$ and $B = \mu_0\lambda = \mu_0\sigma\omega R$.

Magnetic Pressure On A Solenoid:



Consider a small rectangular patch on surface of a solenoid having dimensions $dx \times dl$. Let B_1 be the field created by it in its vicinity and let B_2 be the field due to rest of the solenoid. Now field just below the patch $B = B_1 + B_2 = \mu_0\lambda$ and field just above the patch, $0 = B_2 - B_1$, Thus we can say that:

$$B_1 = B_2 = \mu_0\lambda/2.$$

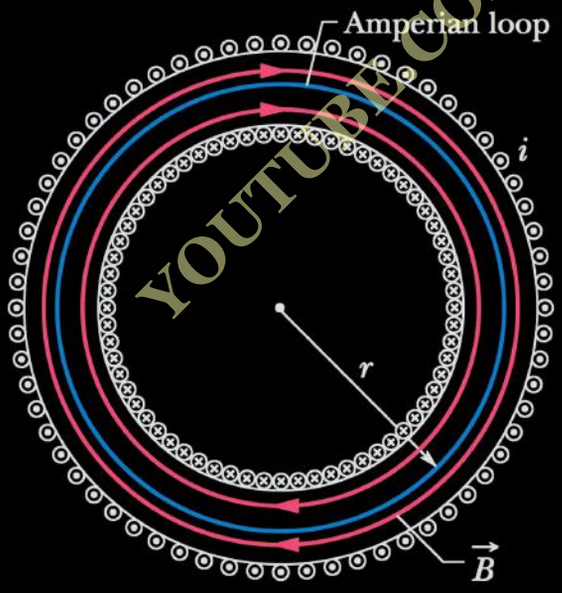
Now force on the patch due to B_2, $dF = i\vec{dl} \times \vec{B}$ i.e $\lambda dx\, dl \times \mu_0\lambda/2$

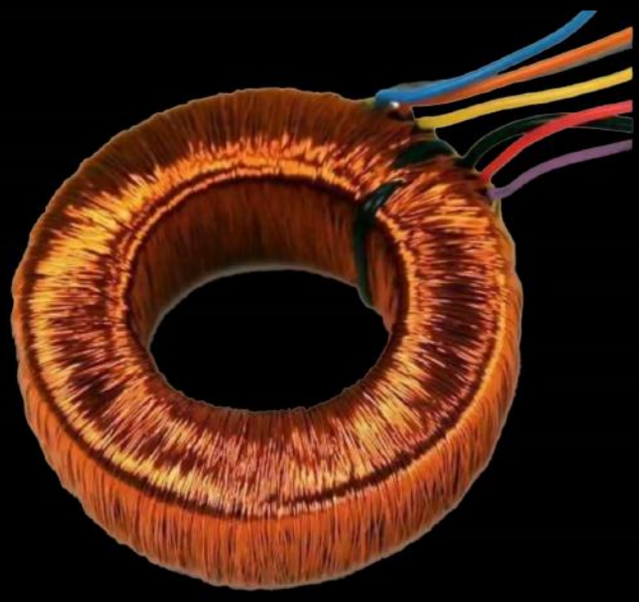
Now Pressure $= \frac{dF}{dA} = \frac{dF}{dx\,dl} = \frac{\mu_0\lambda^2}{2}$, now using $B = \mu_0\lambda$,

$$\boxed{P = B^2/2\mu_0 = \mu_0\lambda^2/2}$$ ← Same formula can be used for spinning shell as well.

This is a repulsive pressure (tries to expand the solenoid).

Magnetic Field Due To A Toroid:





A toroid is a copper coil wound on a solid on a hollow torus (see figure above). Normally the torus has a circular cross section but it could also have a rectangular section. Consider the toroid in the left figure (above) we will find the magnetic field of the toroid using Ampere's law.

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Let n be the number of turns per unit circumference at a radius r (note that n will be a function of r though number of turns per unit angle will be constant. However if cross section radius of toroid is very small in comparison to r, then n may be considered nearly constant). Now applying Ampere's law (Blue loop in figure):

$B \times 2\pi r = \mu_0 (n \times 2\pi r) \times i \Rightarrow B = \mu_0 n i$, thus for a toroid:

$$\boxed{B = \mu_0 n i}$$ ← For points inside the tore.

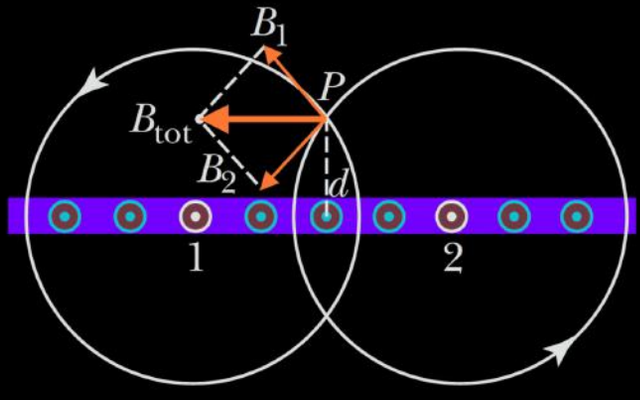
For outside point, the Amperian loop will not enclose any current and one can readily see that outside the toroid field will be zero.

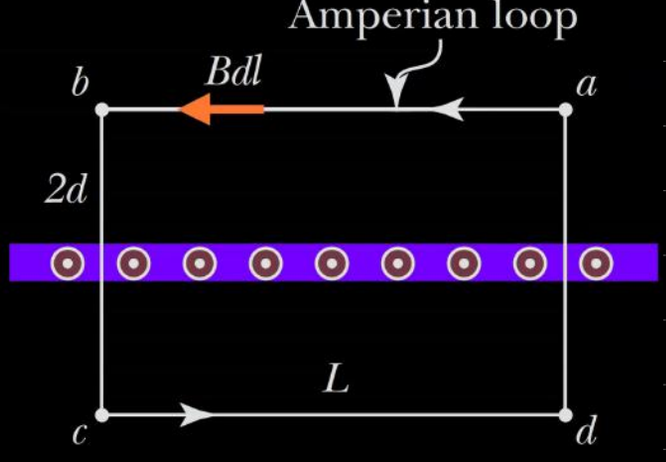
→ In our analysis we assumed perfectly circular turns, however due to advancing of turns, there is an effective tangential current also along the circumference of the toroid causing a field outside the tore to be somewhat like that of a current carrying ring.

Magnetic Field Due To Infinite Sheet Of Current:

Consider an infinite sheet carrying a current λ per unit width coming out of page as shown in the figure. We wish to find out the magnetic field due to this. By considering the elemental currents ① and ② it can be seen that net field at an arbitrary point P must be parallel to the sheet. Now consider an Amperian loop around the portion of the sheet as shown. Now applying Ampere's law:

$B \times L \times 2 = \mu_0 \lambda L \Rightarrow \boxed{B = \frac{\mu_0 \lambda}{2}}$





Method 2: Let σ be the surface charge density of the moving charges, the Electric field contribution because of them:

$E = \sigma/2\epsilon_0$. Now, $\vec{B} = \vec{E} \times \vec{v}/c^2$

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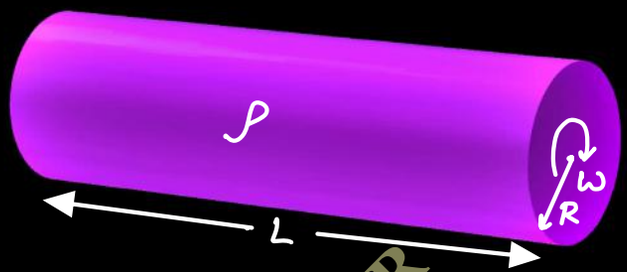
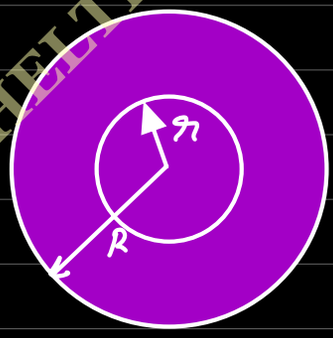
$\Rightarrow B = \frac{\sigma v}{2\epsilon_0 c^2}$, Now $\lambda = \sigma L v dt / L dt = \sigma v$

Thus $B = \frac{\lambda}{2\epsilon_0 c^2} = \mu_0 \lambda / 2$. QED. ($\sigma \to \lambda$, $1/\epsilon_0 \to \mu_0$)

EXAMPLE: A long cylinder having a uniform charge density ρ rotates about its axis with an angular velocity ω. Find the magnetic field as a function of distance r from its axis. The radius of cylinder is R.



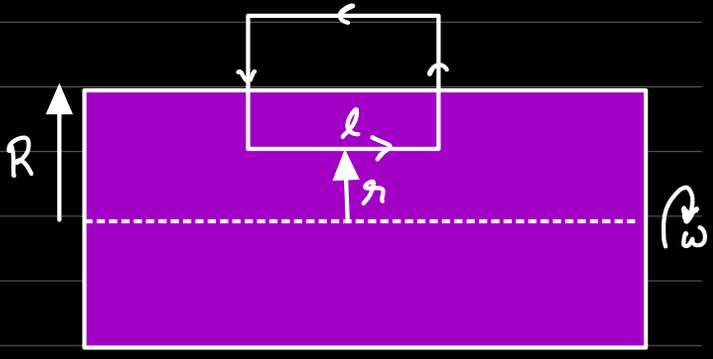
Solution: **Method I:** The system can be visualised as made up of several coaxial solenoidal flows. Recall that a solenoid produces zero field at outside points and a field $\mu_0 \lambda$ for inside points. Thus here, at a radius r, the magnetic field is only due to the charge rotating between radii r and R. Now equivalent λ:



$$\lambda = \frac{\rho \times \pi (R^2 - r^2) \times L \times \omega/2\pi}{L} = \rho \frac{(R^2 - r^2)\omega}{2}$$

Thus $B = \mu_0 \lambda = \mu_0 \rho \omega (R^2 - r^2)/2$ Ans.

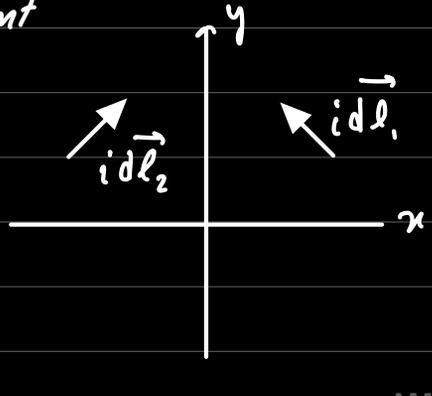
METHOD 2: Consider the amperian loop shown in adjacent figure. Applying Ampere's law:



$$B l = \mu_0 \rho \pi (R^2 - r^2) l \times \omega/2\pi$$

$\Rightarrow B = \mu_0 \rho \omega (R^2 - r^2)/2$ Ans.

EXAMPLE: There are two mirror image current elements in the x-y plane with y-z plane being the mirror. Show that magnetic field anywhere in the bisector plane of the two current elements is perpendicular to the plane.



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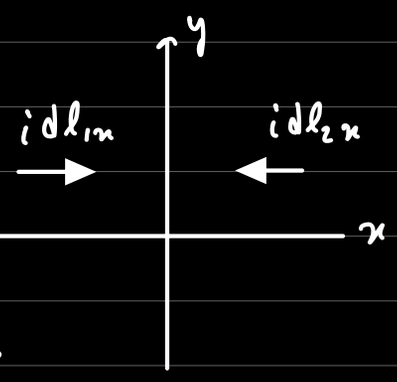
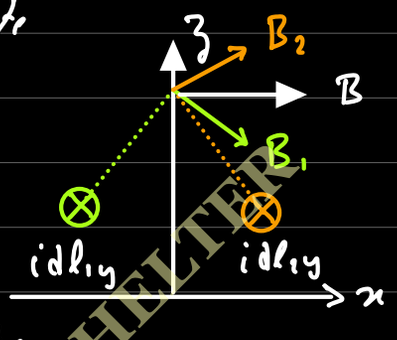
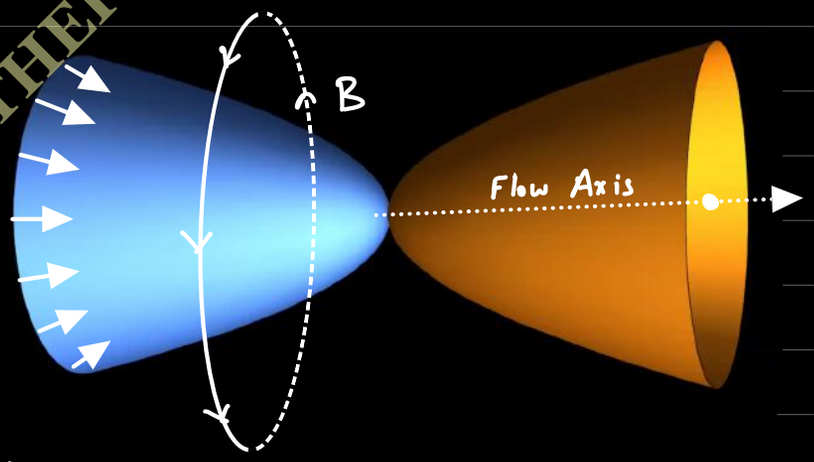
Solution: The current elements can be resolved in to x and y components (or even 3 components if there was any). It can be clearly seen that in the bisector plane i.e y-z plane, the fields due to the two current element x projections cancel out (By right hand thumb rule these elemental components produce equal and opposite components of fields). Now to visualise the field due to y components of current elements, we draw a view in which y axis is going in to the page. Clearly by symmetry we can see that the field is perpendicular to the bisector plane of the mirror image current elements. QED.





Radially Symmetric Axial Current Flow:

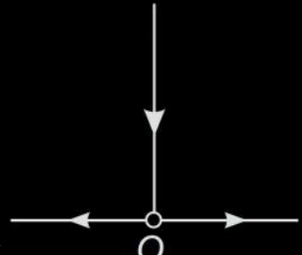
Consider a symmetric surface flow along a double paraboloid, this is an example of radially symmetric axial flow (General flow direction is along axis of the double paraboloid). Whenever this happens, magnetic field lines are coaxial circles centred on the axis of flow.



This is evident from our analysis of previous question. We can imagine a mirror plane passing through any points of space and flow axis and every current element has a mirror image flow in the plane.

Example: Irodov: 3.231

A direct current I flows along a lengthy straight wire. From the point O, the current spreads radially all over an infinite conducting plane perpendicular to the wire. Find the magnetic induction at all points of space.



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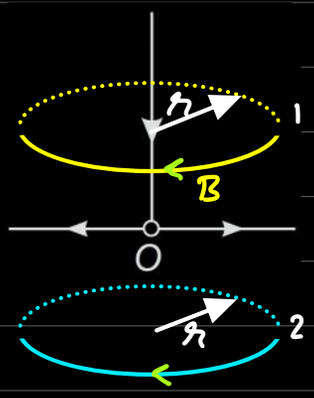
Solution: This is a case of radially symmetric axial flow.

We shall first consider the amperian loop 1 i.e above the infinite plane. Here $B \times 2\pi r = \mu_0 i \Rightarrow B = \mu_0 i/2\pi r$ Ans.

Now for Amperian loop below the plane:

$B \times 2\pi r = \mu_0 \times 0 \Rightarrow B = 0$ Ans.



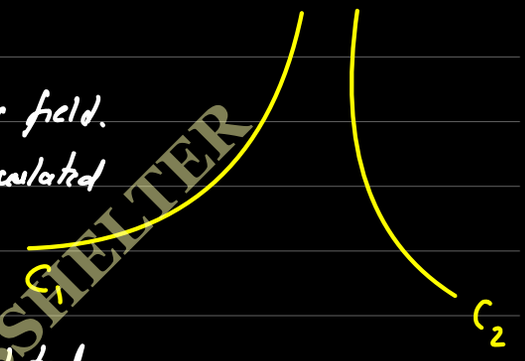
MAGNETIC FIELD INTEGRAL RECIPROCITY (ORIGINAL):

Let C_1 and C_2 be two arbitrary curves now we define following two line integrals of magnetic field.

Case 1: Current i is flowing along C_1 and $\int \vec{B} \cdot d\vec{l}$ is calculated along C_2. Let in this case $\int \vec{B} \cdot d\vec{l} = I_1$

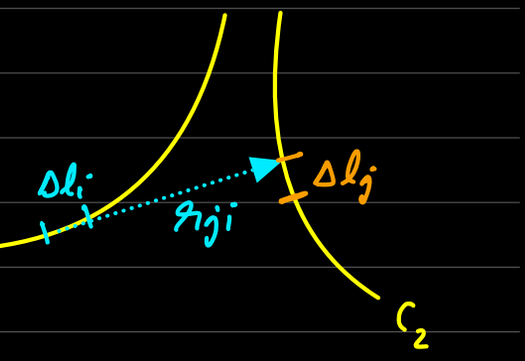
Case 2: Current i is flowing along C_2 and $\oint \vec{B} \cdot d\vec{l}$ is calculated along C_1. Let in this case $\int \vec{B} \cdot d\vec{l} = I_2$ then:

Irrespective of shapes of the curves C_1 and C_2, $|I_1| = |I_2|$.



PROOF: Consider a small element $\Delta\vec{l}_i$ of C_1 and $\Delta\vec{l}_j$ of C_2, for this pair: $\vec{B}_i \cdot \Delta\vec{l}_j$:

$$\vec{B}_i \cdot \Delta\vec{l}_j = \frac{\mu_0}{4\pi} \frac{i(\Delta\vec{l}_i \times \vec{r}_{ji}) \cdot \Delta\vec{l}_j}{(r_{ji})^3} \quad (1)$$



Similarly if current were flowing along Δl_j then $\vec{B}_j \cdot \Delta\vec{l}_i$:

$$\vec{B}_j \cdot \Delta\vec{l}_i = \frac{\mu_0}{4\pi} \frac{i(\Delta\vec{l}_j \times \vec{r}_{ij}) \cdot \Delta\vec{l}_i}{(r_{ij})^3} \quad (2)$$

Now notice the scalar triple products in the numerators of RHS of equations (1) & (2) i.e. $[(\Delta\vec{l}_i \times \vec{r}_{ji}) \cdot \Delta\vec{l}_j) = [(\Delta\vec{l}_j \times \vec{r}_{ij}) \cdot \Delta\vec{l}_i]$

Since these represent volumes of parallelopipes formed by vectors $\Delta\vec{l}_i$, $\Delta\vec{l}_j$ and $\vec{r}_{ij}$. Thus RHS of both equations are equal (other terms are anyway equal). Now for integral over the entire curves,

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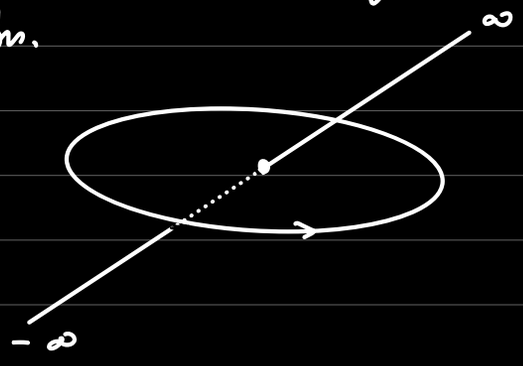
the summation of RHS needs to be done for all i and j. Since each term of the two summations is equal, therefore net summation will also be equal.

EXAMPLE: IRODOV: 3.232:

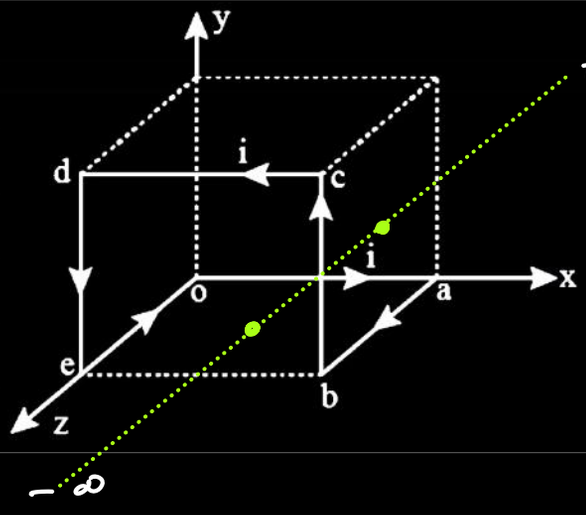
A current I flows along a round loop. Find the integral $\int \vec{B}.d\vec{l}$ along the axis of the loop within the range from $-\infty$ to $+\infty$. Explain the result obtained.

Solution: Here we could either perform the actual integration since we know the formula for magnetic field of a ring, however a smarter method would be to use the reciprocity theorem we just learned in the previous section. i.e. we can flow the current along infinite wire and find $\oint \vec{B}\cdot d\vec{l}$ along the round loop. Thus using Ampere's law we directly have our required result i.e $\oint \vec{B}\cdot d\vec{l} = \mu_0 I$ Ans.

Infact using reciprocity we know that even if the integral were performed along some other axis (other than axis of the loop) such as the one shown, our answer still remains the same i.e. $\int_{-\infty}^{\infty} \vec{B}\cdot d\vec{l} = \mu_0 I$.



Think: What is $\int \vec{B}\cdot d\vec{l}$ along infinite dotted line due to current carrying 3D loop shown in the figure. (The line passes through two face centres of the auxillary cube.



Solution: Using reciprocity, the solution can be worked out directly in the mind. The answer is simply $\mu_0 i$, Just flow the current along infinite wire and use Ampere's law considering the 3D current loop as the Amperian loop.

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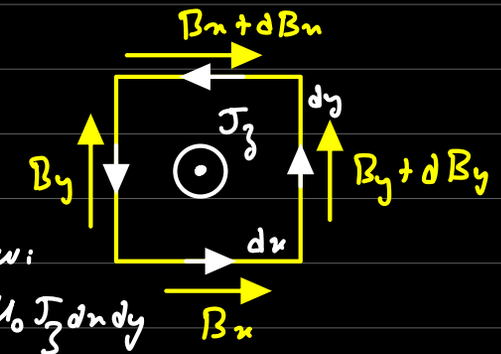
Differential Form Of Ampere's Law:



Consider a small Amperian loop in the xy plane as shown. Let J_z be the z component of local current density. Now applying Ampere's law:

$$B_x\,dx + (B_y + dB_y)\,dy - (B_x + dB_x)\,dx - B_y\,dy = \mu_0 J_z\,dx\,dy$$

Now cancelling terms and dividing by $dx\,dy$:

$$\frac{dB_y}{dx} - \frac{dB_x}{dy} = \mu_0 J_z$$, but since B_x & B_y are multivariable functions, thus here we actually mean partial derivatives. Thus:

$$\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = \mu_0 J_z$$ Similarly:

$$\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} = \mu_0 J_x \quad \text{and} \quad \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} = \mu_0 J_y.$$

Combining all the three equations in to one:

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}}$$ ← Differential form of Ampere's law.

→ If time varying electric fields in space this is modified to Ampere Maxwell law i.e. $\vec{\nabla} \times \vec{B} = \mu_0 \left[\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt} \right]$

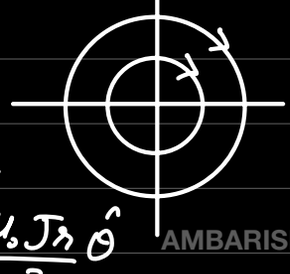
→ In free space devoid of any currents or time varying fields the equation becomes : $\vec{\nabla} \times \vec{B} = 0$. (Just like the curl of conservative forces is zero).

Example: Magnetic field in certain region of space is given as $\vec{B} = y\hat{i} - x\hat{j}$ find the current density in the region.

Solution: Magnetic field is in x-y plane. Now $\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = \mu_0 J_z$

$$\Rightarrow -1 - 1 = \mu_0 J \quad \Rightarrow \quad J = -2/\mu_0.$$

Method 2: Recognise that this field is $-r\hat{\theta}$, Also recall that field due to uniform density conical current is also tangential at an inside point given by $B = \frac{\mu_0 J r}{2}\hat{\theta}$



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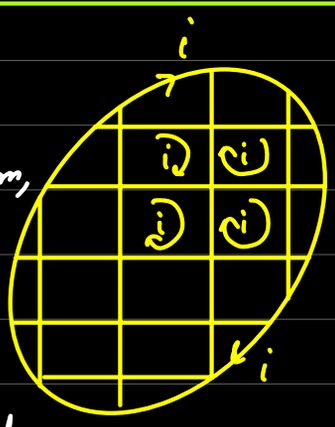
By comparison, $-\pi\hat{\theta} = \mu_0 J \pi/2\ \hat{\theta} \Rightarrow J = -2/\mu_0$ Ans.

Relation Between Current, Flux & Potential Energy Of A Current Loop:

Consider a current carrying loop shown in the figure. Recall that potential energy of such a loop is given by $-\vec{m}\cdot\vec{B}$ if $\vec{B}$ were uniform, however if B is not uniform then the potential energy will have to be calculated as $U = -\Sigma \vec{m}_i \cdot \vec{B}_i$ where $\vec{m}_i$ is the magnetic moment of i^{th} sub-loop and $\vec{B}_i$ the associated magnetic field. (Recall that any finite current loop can be thought of as a collection of infinitesimal rectangular loops each carrying the same current i). Now let $\vec{S}_i$ be the area vector associated with i^{th} sub-loop then:



$$U = -\sum i\vec{S}_i \cdot \vec{B}_i = -i\sum \vec{B}_i \cdot \vec{S}_i = -i\sum \phi_i = -i\phi$$

Thus:

$$\boxed{U = -i\phi}$$

Here ϕ is flux associated with current loop, with Area normal of each subloop being along direction of magnetic moment.

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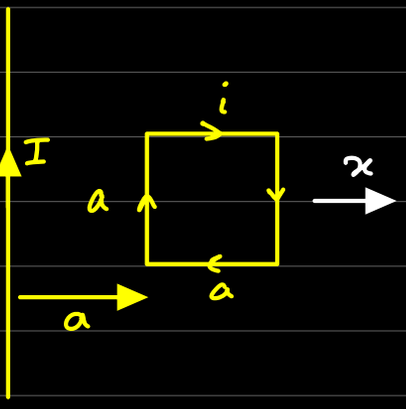
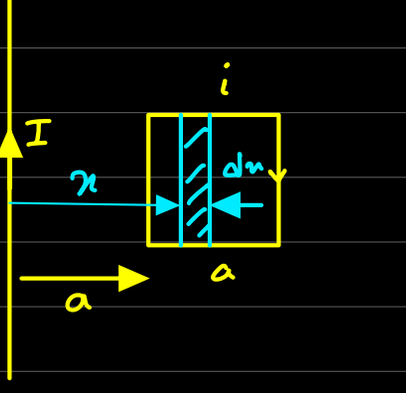
Example: A square loop carrying a constant current i is placed coplanar with an infinite wire carrying a current I. Find (a) flux in this configuration (b) Flux if the loop is translated a distance 'a' along x-axis (c) Work done in the translation process.



Solution: Consider a small element of the loop at a distance x having a thickness dx and area normal inwards. For this element $d\phi = \frac{\mu_0 I}{2\pi x} \times a\,dx$, Now ϕ can be found by integrating this (a) $\phi_i = \frac{\mu_0 I a}{2\pi}\int_a^{2a} \frac{1}{x}\,dx = \frac{\mu_0 I a \ln(2)}{2\pi}$



(b) Similarly we can Integrate here from 2a to 3a

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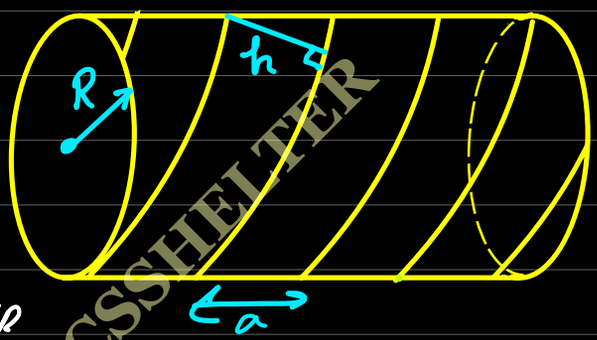
Thus $\phi_f = \frac{\mu_0 I a}{2\pi} \ln(3/2)$ (i) Now $W_{ext} = \Delta U = -i\Delta\phi$

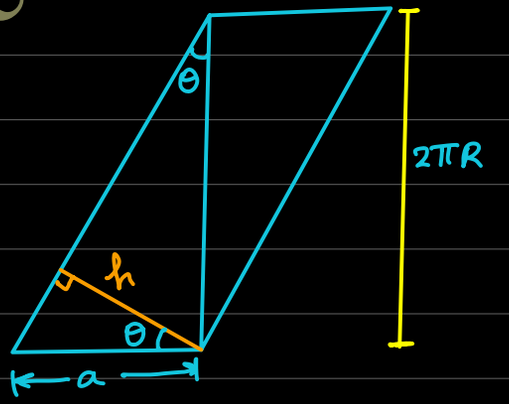
$\Rightarrow \Delta W_{ext} = \frac{\mu_0 I i a}{2\pi} \ln(4/3)$ Ans.

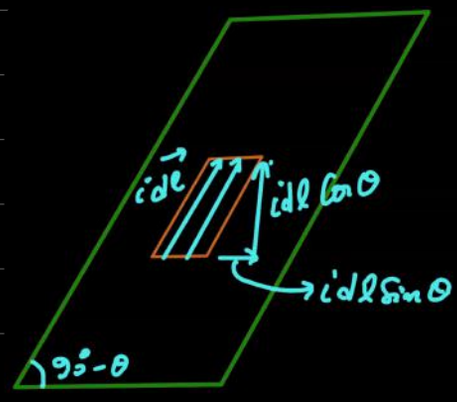
EXAMPLE: IRODOV: 3.238:

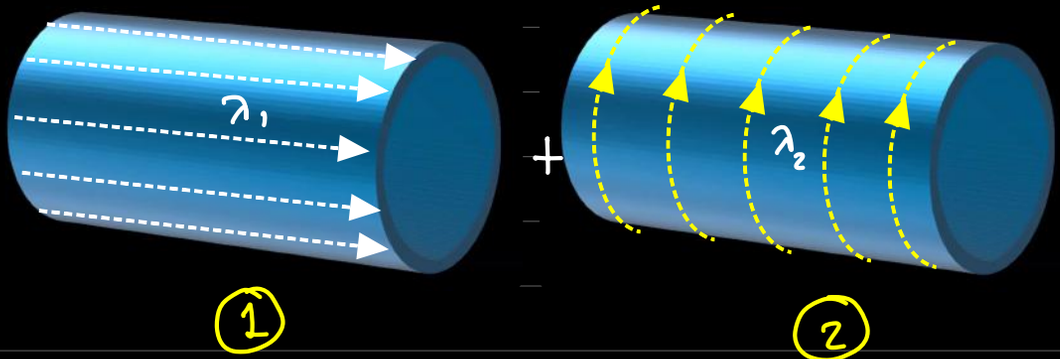
A thin conducting strip of width $h = 2.0$ cm is tightly wound in the shape of a very long coil with cross-section radius $R = 2.5$ cm to make a single-layer straight solenoid. A direct current $I = 5.0$ A flows through the strip. Find the magnetic induction inside and outside the solenoid as a function of the distance r from its axis.



Solution: Let us first unwrap one turn of the ribbon solenoid to understand the geometry. The lower figure shows this unwrapped turn. The vertical projection of unwrap must be equal to $2\pi R$ and horizontal width must be equal to the pitch of the ribbon. For forming a tightly wound coil, left edge of the ribbon must be exactly be above the initial location of the right edge (i.e advance by exactly one pitch). Now $\sin\theta = h/2\pi R$ (1) and $a = h\sec\theta$ (2). Now any current element along the ribbon can be resolved in to axial and tangential components, thus the magnetic effect of ribbon solenoid can be thought of as the superposition of following current geometries:







Now the first component creates the field only outside the solenoid just like an infinite current carrying wire (Recall shell theorem

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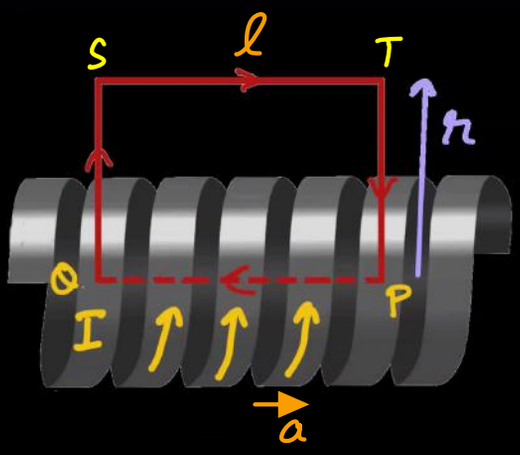
ideal) and the second component creates a field just like that of a solenoid. (In the figure $\lambda_1 \rightarrow$ (current per unit circumference $\lambda_2 \rightarrow$ (current per unit length of solenoid). $\lambda_1 = I/2\pi R$, $\lambda_2 = I/a$

Now we can apply Ampere's law for finding the magnetic fields at inside and outside points.



No of turns per unit length = $1/a$

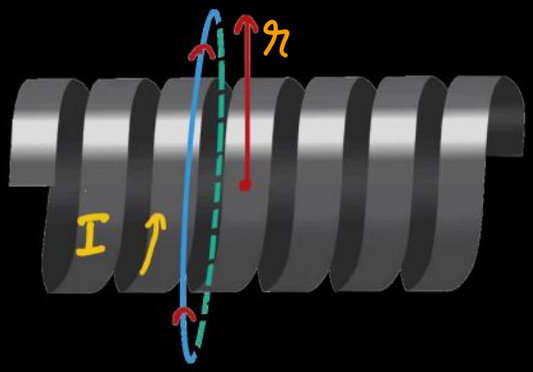
Now Ampere's law, $B \times l = \mu_0 \left(\frac{l}{a}\right) \times I$

$\Rightarrow B = \mu_0 I/a$,

Now $a = h \sec\theta$ and $\sin\theta = h/2\pi R$

$\Rightarrow a = h \times \frac{1}{\sqrt{1-(h/2\pi R)^2}}$

Thus for inside points, $B = \frac{\mu_0 I}{h}\sqrt{1-\left(\frac{h}{2\pi R}\right)^2}$ Ans.



Now for field due to axial flow:

$B \times 2\pi r = \mu_0 I \Rightarrow B = \mu_0 I/2\pi r$ Ans.

(This is the field at outside points, in tangential direction).

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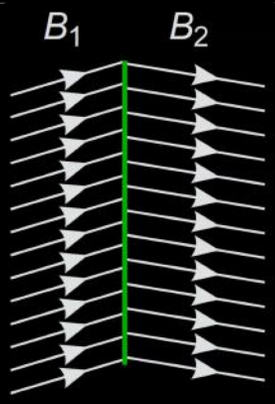
EXAMPLE : IRODOV: 3.260:

A conducting current-carrying plane is placed in an external uniform magnetic field. As a result, the magnetic induction becomes equal to B_1 on one side of the plane and to B_2, on the other. Find the magnetic force acting per unit area of the plane in the shown case, also find the direction of the current.

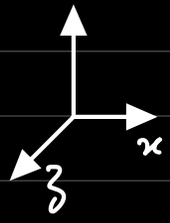


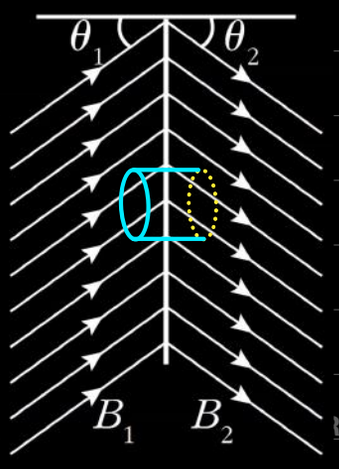
Solution: Let External field be $\vec{B}$ and that due to sheet be $\vec{B'}$ on the right and $-\vec{B'}$ on the left, then:

$$\vec{B} + \vec{B'} = \vec{B}_2 \quad \text{and}$$

$$\vec{B} - \vec{B'} = \vec{B}_1$$

$$\Rightarrow \vec{B} = \frac{\vec{B}_1 + \vec{B}_2}{2} \quad \text{and} \quad \vec{B'} = \frac{\vec{B}_2 - \vec{B}_1}{2}$$

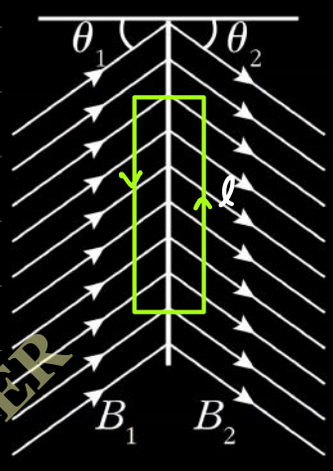




Now assume a small Gaussian Pill Box near the sheet as shown. Using Gauss law for magnetism: $B_2 S\cos\theta_2 - B_1 S\cos\theta_1 = 0 \Rightarrow B_1\cos\theta_1 = B_2\cos\theta_2$.

Thus $\vec{B}$ does not have any x component and has only y component, which must be caused by a current flowing in z direction.

Let current be unit width of strip be $\lambda\hat{k}$. Now applying Ampere's law (see adjacent figure):



$$-B_1 l\sin\theta_1 - B_2 l\sin\theta_2 = \mu_0 \lambda l$$

$$\Rightarrow \vec{\lambda} = -(B_1\sin\theta_1 + B_2\sin\theta_2)\hat{k}/\mu_0$$

Now let inward depth of sheet be w. Now force on the sheet is given as: $\vec{F} = i\vec{l}\times\vec{B}$, in our case,

$$\vec{F} = -(B_1\sin\theta_1 + B_2\sin\theta_2)\frac{lw}{\mu_0}\hat{k}\times\left[B_1\cos\theta_1\hat{i} + B_2\cos\theta_2\hat{i} + B_1\sin\theta_1\hat{j} - B_2\sin\theta_2\hat{j}\right]/2$$

Thus x component of force:

$$F_x = \frac{A}{2\mu_0}\left[B_1^2\sin^2\theta_1 - B_2^2\sin^2\theta_2\right]$$

Also $B_1^2\sin^2\theta_1 = B_1^2 - B_1^2\cos^2\theta_1$ similarly $B_2^2\sin^2\theta_2 = B_2^2 - B_2^2\cos^2\theta_2$

But $B_1\cos\theta_1 = B_2\cos\theta_2$ Thus:

$$F_x = \frac{A}{2\mu_0}\left[B_1^2 - B_2^2\right] \Rightarrow \text{Pressure: } P = \frac{B_1^2 - B_2^2}{2\mu_0}$$

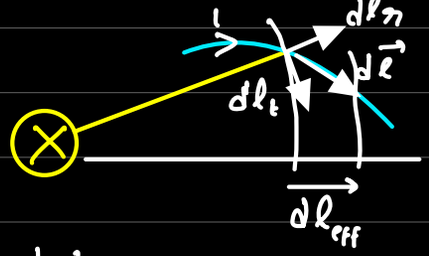
Now y component of force: $F_y = -\frac{A}{\mu_0}(B_1\sin\theta_1 + B_2\sin\theta_2)(B_1\cos\theta_1 + B_2\cos\theta_2)$

Thus shear stress $\sigma_{sy} = (B_1\sin\theta_1 + B_2\sin\theta_2)(B_1\cos\theta_1 + B_2\cos\theta_2)/2\mu_0$ Ans

→ (Inodov's Answer hey has ignored the shearing force).

Magnetic Force On A Conductor In A Plane ⊥ To Infinite Wire Of Current:

Consider an arbitrary wire carrying a current i in a plane ⊥ to infinite wire of current I. A small $d\vec{l}$ of the arbitrary wire can be resolved in to tangential and radial components. Now magnetic force on tangential component of $i\vec{dl}$ is zero because magnetic field lines are tangential, the force is experience only by $i\vec{dl}_{\perp}$



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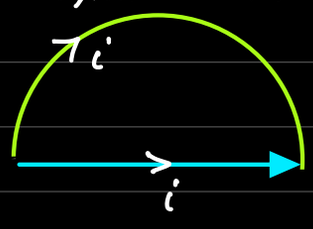
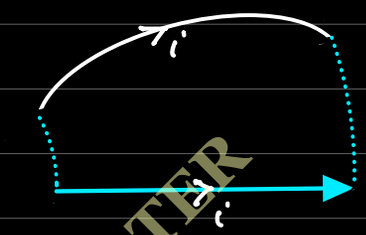
which is same as the force experience by its radial projection on a horizontal line passing through current wire i.e $i\vec{dl}_{eff}$. Thus any arbitrary wire in this plane can be just replaced by its radial projection on line passing through infinite wire.



For instance the semicircular wire can be simply replaced by its diameter, for purpose of calculating the force.

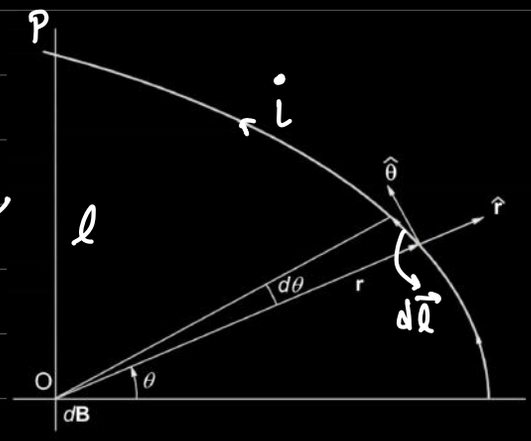
Similarly the elliptical segment (white) can be replaced by its radial projection, (turquoise line).



→ This also implies that force on a closed loop in this plane must be zero as its radial projection is zero. (Even though torque will be non-zero in general).

Magnetic Field At Focus Of Current Carrying Conic Section:

Recall that polar equation of any conic section is given as $r = \frac{l}{1+e\cos\theta}$ where l is the semi-latus rectum and e is the eccentricity, r is the position vector from the focus.



Now any current element $i\vec{dl}$ can be resolved in to radial and tangential components i.e $i\vec{dl} = i[r d\theta\hat{\theta} + dr\hat{r}]$

However radial component creates no field at the focus. Now field due to tangential component using Biot & Savart's law:

$$dB = \frac{\mu_0}{4\pi}\frac{i\,r\,d\theta \times r}{r^3} = \frac{\mu_0 i\,d\theta}{4\pi r} = \frac{\mu_0 i}{4\pi l}(1+e\cos\theta)\,d\theta$$

Now $B = \int dB = \frac{\mu_0 i}{4\pi l}\int_0^{2\pi}(1+e\cos\theta)\,d\theta = \frac{\mu_0 i}{2l}$

(Latus Rectum is the focal chord $\perp$ to the direction $\theta = 0°$).

→ Magnetic field is independent of eccentricity. For conics with same latus rectum, magnetic field is same (for same current).

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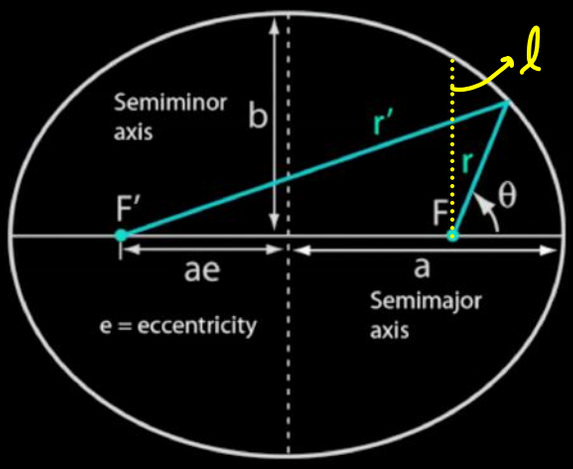
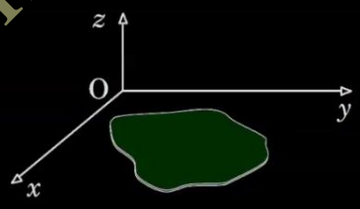
→ As a ready reference, parameters of ellipse are shown. Here l is the semi-latus rectum. $l = b^2/a$.
[Which is also radius of curvature at end point of major axis].



EXAMPLE: PATHFINDER BUILD-1

A lamina made of an insulating material carrying some amount of charge is placed in the x-y plane of a coordinate system as shown in the figure. Electrostatic potential at the origin O due to the charges on the lamina is V_0. Now the lamina is made to rotate with a constant angular velocity ω about the z-axis. Find magnetic field at the origin created by the moving charges of the lamina.



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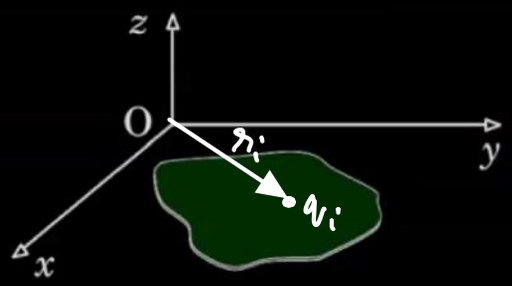
Solution: Consider i^{th} charged particle of the lamina having a charge q_i, due to this magnetic field at the origin:

$$\vec{B}_i = \frac{\mu_0}{4\pi} \frac{q_i (\vec{\omega} \times (-\vec{r}_i))}{r_i^3}$$



$$\Rightarrow \vec{B}_i = -\frac{\mu_0}{4\pi r_i^3} q_i (\vec{\omega} \times \vec{r}_i) \times \vec{r}_i = \frac{\mu_0}{4\pi} \frac{q_i}{r_i^3} [\vec{\omega}(\vec{r}_i \cdot \vec{r}_i) - \vec{r}_i (\vec{\omega} \cdot \vec{r}_i)]$$

But $\vec{\omega} \cdot \vec{r}_i = 0$ thus: $\vec{B}_i = \frac{\mu_0}{4\pi} \frac{q_i}{r_i} \vec{\omega} \Rightarrow \vec{B} = \frac{\mu_0 \vec{\omega}}{4\pi} \Sigma \left(\frac{q_i}{r_i}\right)$

Also potential at origin due to i^{th} charge, $V_i = \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i}$, thus,

$$\vec{B} = \frac{\mu_0 \vec{\omega}}{4\pi} \cdot 4\pi\epsilon_0 \Sigma V_i = \frac{V_0}{c^2} \vec{\omega} \text{ Ans.}$$

EXAMPLE: PATHFINDER: CHALLENGE-2:

Consider a long current carrying cylindrical conductor of radius r. Current density j inside the conductor is uniform over its cross-section. Deduce suitable expression for force of interaction per unit length between two halves that are obtained by dividing the conductor by a plane containing the axis of the conductor.

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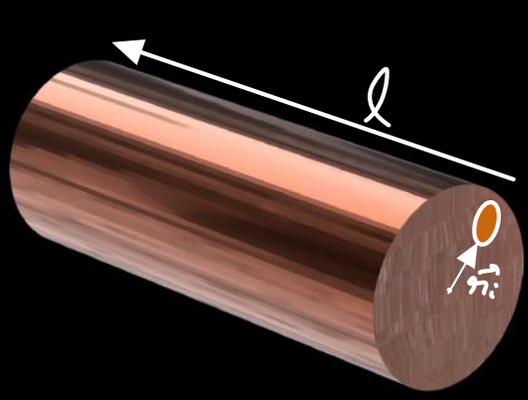
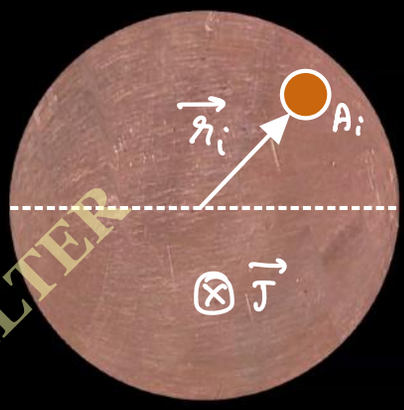
Solution: Recall that magnetic field inside a cylinder carrying uniform axial current density is given as $\vec{B} = \mu_0 \vec{J} \times \vec{R}/2$. Now consider a fiber of cross section area A_i at position vector $\vec{r}_i$ as shown in the figure. Force experienced by this fiber:



$$\vec{F}_i = i\vec{l} \times \vec{B}_i = J A_i \vec{l} \times \vec{B}_i =$$



$$\Rightarrow \vec{F}_i = A_i l \vec{J} \times (\mu_0 \vec{J} \times \vec{r}_i/2)$$

$$\Rightarrow \vec{F}_i = \mu_0 A_i l/2 \left[\vec{J}[\vec{J}\cdot\vec{r}_i] - \vec{r}_i[\vec{J}\cdot\vec{J}]\right]$$

$$\Rightarrow \vec{F}_i = \frac{\mu_0 A_i l}{2}\left[-\vec{r}_i J^2\right] = \frac{-\mu_0 J^2 l}{2} A_i \vec{r}_i$$

Now taking Summation on upper semi-cylinder, $\vec{F} = \frac{-\mu_0 J^2 l}{2} \frac{\sum A_i \vec{r}_i}{A} \times A$

where $A = \pi R^2/2$, thus $\vec{F} = -\frac{\mu_0 J^2 l}{2} \times \frac{\pi R^2}{2} \times \frac{4R}{3\pi}(\hat{y})$

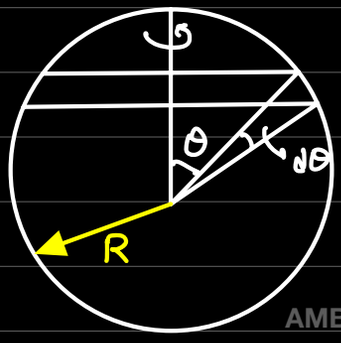
$$\Rightarrow \vec{F} = -\frac{\mu_0 J^2 l R^3}{3}\hat{y}$$ Thus magnitude of force per unit length:

$$\frac{F}{l} = \mu_0 J^2 R^3/3 \text{ Ans.}$$

EXAMPLE: PATHFINDER: BUILD-4:

An isolated conducting sphere of radius R maintained at a potential V is being rotated at a constant angular velocity ω about one of its diameter. Find magnetic dipole moment $\vec{m}$ of the sphere and magnetic induction $\vec{B}$ at its centre.

Solution: We can find magnetic field by integrating field of ring shaped elements.



Recall field due to a ring,

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$$B = \frac{\mu_0 i}{2}\frac{R^2}{(z^2+r^2)^{3/2}}$$

For our case, $dI = \left(\frac{Q}{4\pi R^2} \times R\,d\theta \times 2\pi R\sin\theta\right).\frac{\omega}{2\pi}$

$r = R\sin\theta$ and $(z^2+r^2) = R^2$.

Substituting all the values,

$$dB = \frac{\mu_0 Q\omega}{8\pi R}\sin^3\theta\,d\theta \Rightarrow B = \frac{\mu_0 Q\omega}{8\pi R}\int_0^{\pi}\sin^3\theta\,d\theta$$

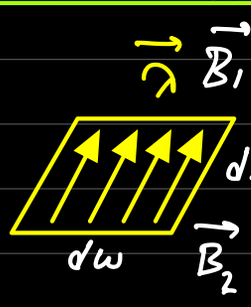
Also $V = \frac{Q}{4\pi\epsilon_0 R} \Rightarrow Q = 4\pi\epsilon_0 RV$, substituting we get

$$\vec{B} = \frac{2\epsilon_0\mu_0 V\vec{\omega}}{3}$$ Ans. ← This is also the field throughout the sphere.

Now magnetic moment can be found using concept of gyromagnetic ratio i.e $\frac{m}{L} = \frac{q}{2M}$, thus:

$$\vec{m} = \frac{4\pi\epsilon_0 R^3 V}{3}\vec{\omega}$$ Ans.

Magnetic Pressure And Shear Stress On Arbitrary Current Carrying Patch:



Let $\vec{\lambda}$ be the current per unit length of a patch and let $\vec{B}_1$ and $\vec{B}_2$ be the magnetic fields just above and below it, further let field due to this element be $\vec{B}_0$ and $-\vec{B}_0$ just above and below it and $\vec{B}_{ext}$ be the field due to rest of the universe then:

$$\vec{B}_{ext} + \vec{B}_0 = \vec{B}_1 \text{ and } \vec{B}_{ext} - \vec{B}_0 = \vec{B}_2$$

$$\Rightarrow \vec{B}_{ext} = (\vec{B}_1 + \vec{B}_2)/2, \text{ Now } d\vec{F} = (\vec{\lambda}\,dw)\,dl \times \vec{B}_{ext}$$

$$\Rightarrow \text{Force per unit area } \frac{d\vec{F}}{dA} = \vec{\lambda} \times \vec{B}_{ext} = \frac{\vec{\lambda}}{2} \times (\vec{B}_1 + \vec{B}_2)$$

Now let $\hat{A}$ be the area normal unit vector then pressure:

$$P = \left[\frac{\vec{\lambda}}{2} \times (\vec{B}_1 + \vec{B}_2)\right]\cdot \hat{n}$$, Shear stress can be found by subtracting normal force from net force, thus, shear stress:

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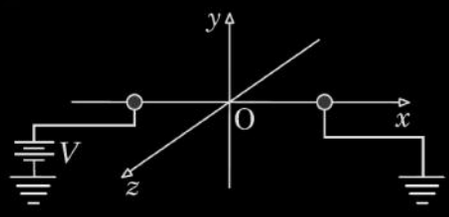
$$\sigma_s = \left| \left[\left(\frac{\vec{\lambda}}{2} \times (\vec{B}_1 + \vec{B}_2) \right) - \left(\frac{\vec{\lambda}}{2} \times (\vec{B}_1 + \vec{B}_2) \cdot \hat{n} \right] \hat{n} \right] \right|$$

EXAMPLE : PATHFINDER : CHECK 2 :

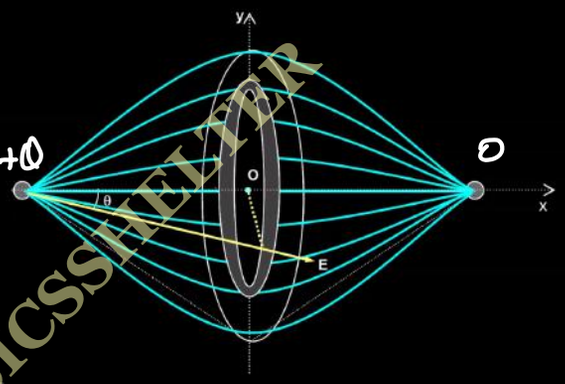
Two identical metallic spheres each of radius r are fixed with their centres at points $(d, 0, 0)$ and $(-d, 0, 0)$ in a homogeneous medium of conductivity σ. Here $d >> r$. One of them is maintained at potential V and the other one is grounded as shown in the figure. Calculate magnetic flux density at a point that is in the y-z plane at a distance R from the origin O.



Solution: Recall that $di = \vec{J} \cdot d\vec{S} = \sigma \vec{E} \cdot d\vec{S}$

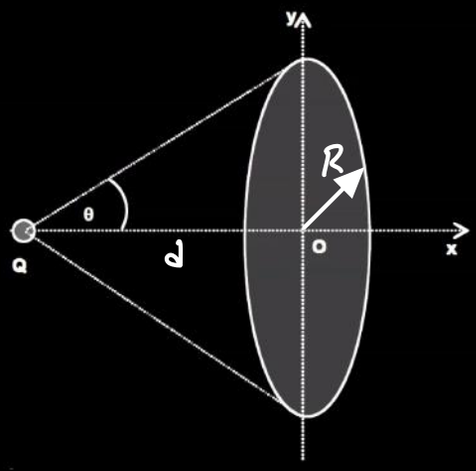
$\Rightarrow di = \sigma d\phi$ where σ is conductivity

This means that current through any Amperian Surface can be simply found using flux



Also recall that flux due to a point charge on axis of a disk is $q/2\epsilon_0 (1 - \cos\theta)$ where θ is the semi-vertex angle subtended by the charge on the disc.

Thus: $B \times 2\pi R = \mu_0 \int \sigma \vec{E} \cdot d\vec{S} = \mu_0 \sigma \int \vec{E} \cdot d\vec{S}$



Also $\frac{Q}{4\pi\epsilon_0 R} = V \Rightarrow Q = 4\pi\epsilon_0 R V$

Now $\cos\theta = d/\sqrt{R^2 + d^2}$, putting in values,

$$B \times 2\pi R = \mu_0 \sigma \times 2\pi R V \left(1 - \frac{d}{\sqrt{R^2 + d^2}} \right)$$

$$\Rightarrow B = \frac{\mu_0 \sigma R V d}{R} \left(1 - \frac{d}{\sqrt{R^2 + d^2}} \right) \text{ Ans.}$$

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